

Ultra-high-throughput mapping of genetic design space

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Massively parallel genetic screens have been used to map sequence-to-function relationships for a variety of genetic elements^{1–5}. However, as these approaches interrogate only short sequences, it remains challenging to perform high-throughput assays on constructs containing combinations of multiple sequence elements arranged across multi-kb length scales. Overcoming this barrier could accelerate synthetic biology; by screening diverse gene circuit designs and learning ‘composition to function’ mappings, genetic part composability rules could be revealed, enabling rapid identification of behaviour-optimized design variants^{6,7}. Here we introduce CLASSIC (combining long- and short-range sequencing to investigate genetic complexity), a genetic screening platform that combines long- and short-read next-generation sequencing (NGS) modalities to quantitatively assess pools of constructs of arbitrary length containing diverse genetic part compositions. We show that CLASSIC can measure expression profiles of over 10⁵ gene circuit designs (from 5–20 kb) in a single experiment in human cells. The resulting datasets can be used to train machine-learning models that accurately predict circuit behaviour across expansive circuit design landscapes, revealing part composability rules that govern circuit performance. Our study shows that, by expanding the throughput of each design–build–test–learn cycle, CLASSIC enhances the pace and scale of synthetic biology and establishes an experimental basis for data-driven design of complex genetic systems.

Synthetic gene circuits are constructed by assembling DNA-encoded genetic parts into multigene programs that perform computational tasks in living cells^{8–10}. Over the past two decades, gene circuits have emerged as important models for understanding native gene regulation¹¹ and have been used to create powerful biotechnologies by enabling user-defined control over cellular behaviour^{12–14}. Despite this progress, the design of quantitatively precise circuit behaviour remains challenging. Regulatory interactions within a circuit must be carefully tuned, often through iterative design–build–test–learn (DBTL) cycles, before part compositions that support a desired circuit behaviour are identified¹⁵. Moreover, as genetic parts must work in close physical proximity within a crowded intracellular environment¹⁶, incidental molecular coupling can occur between parts and with host cell regulatory machinery^{17,18}. These context-dependent interactions can confound circuit design approaches that use biophysical and mechanistic modelling frameworks, extending the number of DBTL cycles required to achieve a target behaviour.

One potential strategy to accelerate gene circuit engineering is to expand the number of circuits tested per cycle by performing high-throughput functional screens on pooled circuit libraries.

By measuring large collections of circuits in a single experiment, such an approach could enable rapid profiling of complex circuit design spaces to identify part compositions with desired quantitative behaviours. In contrast to traditional approaches, large-scale circuit activity data could also facilitate training of ML/artificial intelligence models capable of inferring context-specific part function and more accurately predicting the behaviour of untested circuit designs. Short-read next-generation sequencing (NGS)-based high-throughput screens^{19–21} have been used to generate detailed sequence-to-function mappings for multiple classes of discrete genetic parts, including promoters^{1,4,22–24}, terminators and 3' untranslated regions (UTRs)^{25,26}, transcription factors (TFs)^{27,28}, nucleic acid switches³ and receptors⁵. However, high-depth functional profiling of libraries of DNA constructs long enough (>1 kb) to encode entire multipart circuits remains costly and technically challenging. Methods have emerged for multiplexed analysis of long constructs by appending short barcode index sequences for amplicon-based readout by short-read NGS²⁹. This approach has been used in array formats, in which each construct is assembled and barcoded separately, or through nested assembly schemes^{29,30}, limiting library design flexibility and throughput. More recently, random

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barcoding and indexing through long-read single-molecule real-time sequencing (PacBio) has enabled characterization of libraries ranging from 10^2 to 10^4 constructs, mostly for individual open reading frames (ORFs) of up to 6 kb in length^{31,32}.

To perform indexed multiplexing for construct libraries containing compositions of multiple genetic parts at larger length scales, we devised an approach combining long-read Nanopore and short-read Illumina NGS to analyse construct libraries generated through pooled part assembly (Fig. 1a). Our assembly scheme incorporates structured barcodes that can be used to create a construct composition-to-barcode index by Nanopore sequencing. Libraries are then introduced into cells, binned based on functional phenotype, and analysed by short-read (Illumina) amplicon sequencing to produce a barcode-to-phenotype index. By comparing the two indices, construct composition can be matched to phenotype. This approach, termed CLASSIC, enables collection of high-depth phenotypic expression data for large genetic part composition libraries ($>10^5$) of arbitrary length using standard phenotypic selection or flow sorting experiments.

To configure CLASSIC to quantitatively profile diverse construct libraries in human cells (Fig. 1b and Supplementary Fig. 1), we implemented a custom hierarchical Golden Gate³³ cloning scheme that generates library diversity through a series of pooled DNA assembly steps (Supplementary Figs. 2–4). Diversified pools of 5', coding and 3' gene elements are first generated through assembly of input part fragments (level 0 to 1), and subsequently combined to create single-gene expression unit (EU) pools (level 1 to 2) (Supplementary Fig. 2). The EU pools are combined with a BFP EU plasmid pool expressing a transcript containing semi-degenerate barcode sequences in the 3' UTR (Supplementary Fig. 5), yielding barcoded multi-EU libraries (level 2 to 3). After Nanopore sequencing and data analysis (Methods and Supplementary Fig. 6), the libraries are genomically integrated at single copy into a HEK293T cell line harbouring a custom landing-pad cassette (HEK-LP)³⁴ (Methods and Supplementary Fig. 7). Library-integrated cells are then flow-sorted based on circuit expression output, followed by Illumina NGS to quantify bin distributions of circuit-associated barcodes^{19,20} (Methods and Supplementary Fig. 8). We validated this workflow with a small ($n = 384$) EU library containing shuffled regulatory elements (promoter, Kozak sequence and terminator³⁵) (Supplementary Fig. 9), demonstrating that our scheme yields complete, part-balanced library coverage (Supplementary Fig. 10a–c) and proportional assortment of unique barcodes (Supplementary Fig. 10d,e). HEK-LP library integration (Supplementary Fig. 11a–c) and flow-seq measurements were compared with flow cytometry measurements of randomly isolated, clonally expanded variants (Supplementary Fig. 12a,b and Supplementary Table 2). These ground-truth measurements showed excellent agreement with corresponding CLASSIC-derived values (mean absolute error (MAE) = 0.07), and demonstrated that measurement precision is enhanced by mapping multiple unique barcodes to each composition (Supplementary Fig. 12c). We also collected replicate libraries to demonstrate that CLASSIC is technically ($r^2 = 0.99$) (Supplementary Fig. 12e) and biologically ($r^2 = 0.97$) reproducible (Supplementary Fig. 12d), and that aggregating expression data from multiple barcodes leads to higher measurement accuracy (Supplementary Fig. 12e,f). Finally, we showed that the data enable quantification of relative part contributions to expression magnitude across design space (Supplementary Fig. 13a,b) that can be accurately predicted by ML models such as random forests (Supplementary Fig. 14).

To evaluate whether CLASSIC enables experimental profiling of complex, multigene circuit design spaces, we constructed a library of small-molecule-inducible, single-input transcriptional switches comprising over 10^5 potential unique compositions. Circuits in this library contain two EUs (Fig. 2a): one encoding a constitutively expressed synthetic Cys2-His2 zinc-finger (ZF)-based TF (synTF)^{36–38} with appended transcriptional activation (TA) domains, and the other encoding a reporter gene containing cognate synTF binding motifs (BMs) upstream of a minimal promoter driving eGFP expression³⁶. Circuit induction occurs

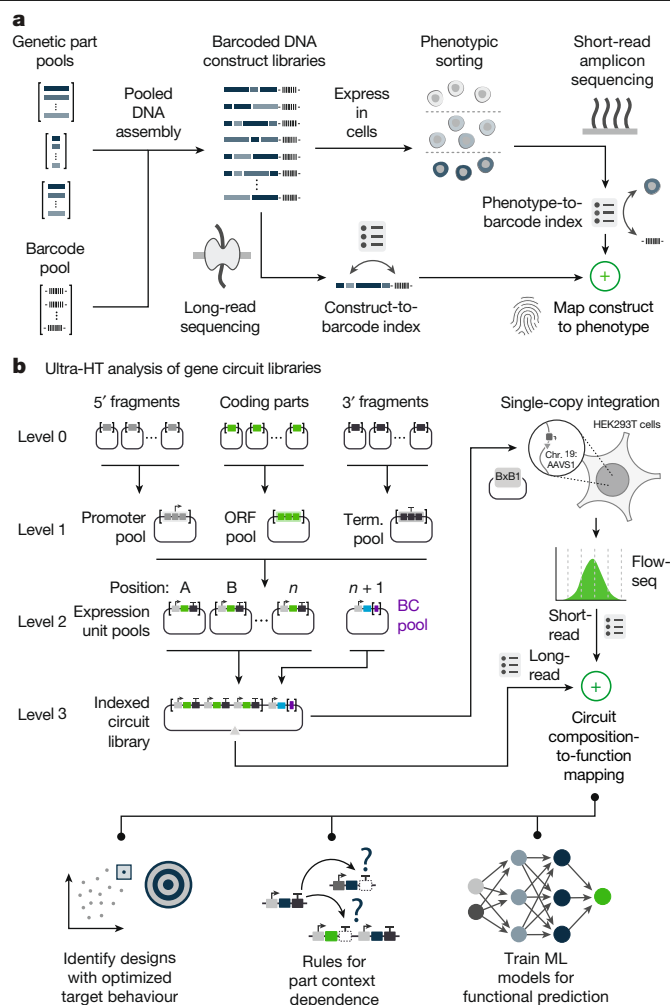


Fig. 1 | Using CLASSIC to systematically map the design space of complex genetic programs. **a**, Overview of CLASSIC. Pooled assembly of genetic parts with DNA barcode sequences yields libraries of barcode-indexed constructs of arbitrary length and complexity. Long-read Nanopore sequencing is used to create an index matching construct composition to an associated barcode. In parallel, libraries introduced into cells undergo sorting or selection to bin expression phenotypes. Barcode amplicons generated for each bin are subjected to short-read NGS to quantify the expression phenotype, which is then mapped to construct composition through barcode indexes. **b**, Application of CLASSIC to profile a synthetic gene circuit design space. Hierarchical Golden Gate assembly is used to compose libraries of multi-EU circuits with combinatorially varied part compositions and circuit designs. Sequence fragment pools for different part categories from level 0 are combined to yield level 1 pools of promoters, ORFs and terminators (term.), which are then combined to yield level 2 EUs (square brackets: fragment/part pools). Barcode (BC) pools in BFP EUs are combined with library EU pools to create indexed multi-EU circuit libraries (level 3) that are integrated into HEK-LP cells at the AAVS1 locus in chromosome (chr.) 19 through expression of the BxB1 recombinase (top right). Library analysis performed using a combination of Nanopore and flow-seq yields composition-to-function mapping.

after addition of 4-hydroxytamoxifen (4-OHT), which binds to a mutant version of the human oestrogen receptor (ERT2)³⁹ appended to the synTF, facilitating cytoplasm-to-nuclear translocation to activate reporter transcription. As we and others have demonstrated^{36,40–42}, identification of designs optimized for high-fold-change (HFC) (Fig. 2a (right)) expression can be challenging in mammalian systems; robust, inducer-dependent transgene expression requires basal and induced expression levels to be respectively minimized and maximized through tuning of locus transcriptional regulatory features and synTF molecular properties.

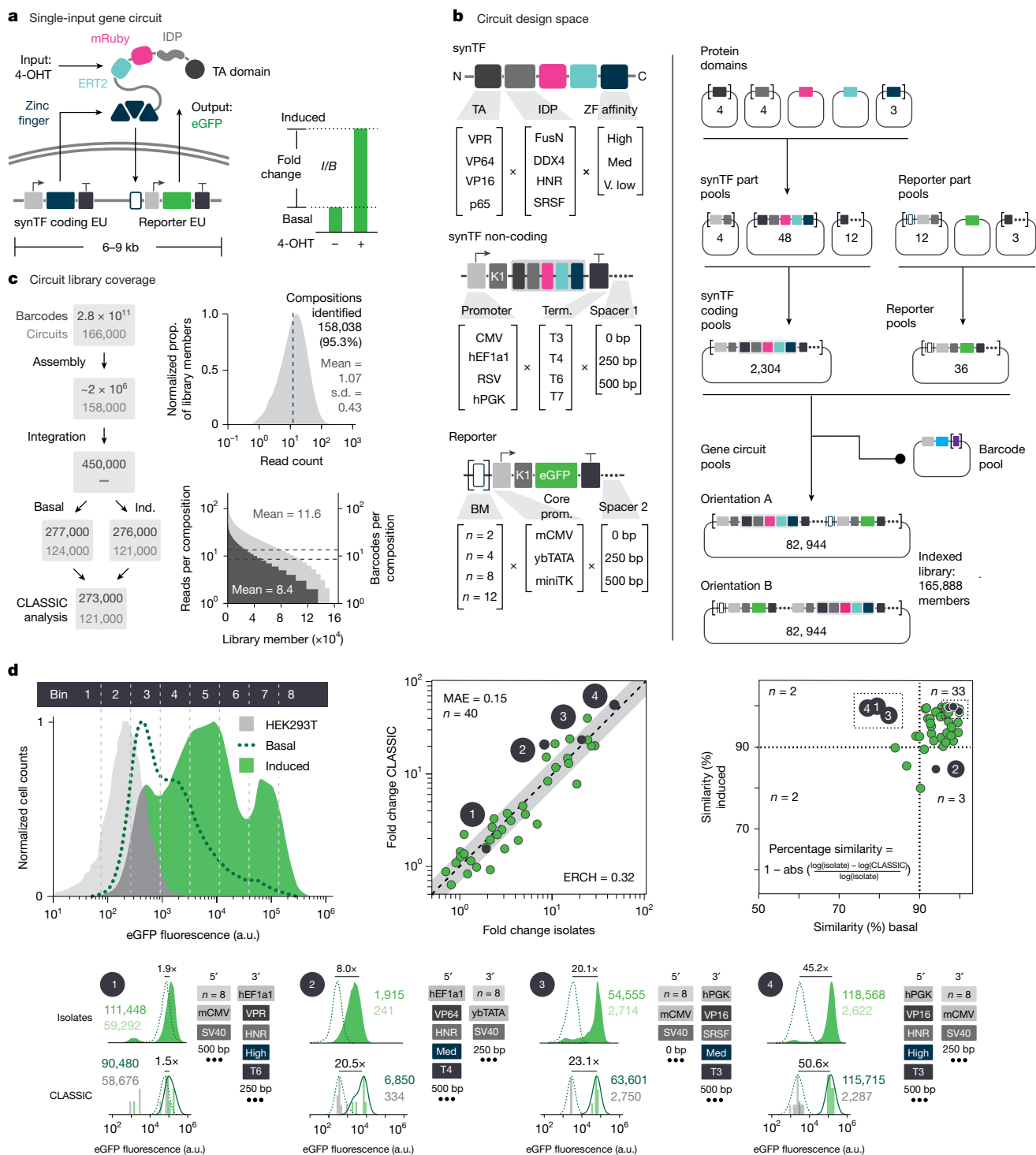


Fig. 2 | Using CLASSIC to quantitatively profile a synthetic gene circuit design landscape. **a**, Circuit diagram of the inducible synTF. Left, the circuit contains a synTF-coding unit and an eGFP reporter unit. Without an inducer, the synTF is cytoplasmic. Addition of 4-OHT (input) triggers nuclear translocation and eGFP activation (output). Right, the expression FC is the ratio of induced (*I*) to basal (*B*) expression levels. **b**, Circuit design space. Left, synTF diversity: 4 TA domains, 4 IDPs and 3 ZF affinity mutants. High, wild type; med, ΔC ; v. low, $\Delta C 6\times$. synTF-coding EU diversity: 4 constitutive promoters (prom.) and 4 terminators. Reporter EU diversity: 4 BM number variants, 3 core promoters and a constant terminator. Both EUs have 3' spacing sequences of 0 bp, 250 bp or 500 bp. Right, circuits are combinatorially assembled from input parts to generate 165,888 possible variants. **c**, Balance of circuit library assembly and indexing. Left, CLASSIC recovered around 121,000 out of 166,000 compositions (73%). Compositions are shown in light grey and barcodes are shown in dark grey. Right, Nanopore

sequencing of the library reveals library composition (mean $\pm$ s.d. computed on a $\log_{10}$ scale) (top) and composition/barcode balance (bottom) (reads per composition (light grey), unique barcodes per composition (dark grey); data are plotted in rank order of reads per composition). Ind., induced. Numbers are rounded to the nearest thousandth. **d**, Circuit library sorting and measurement. Top right, induced (solid green) and basal (dotted green line) eGFP expression of the library with flow sorting bins (vertical grey dotted lines); grey histogram shows HEK293T cells. Top middle, FC values for 40 clonal isolates versus CLASSIC measurements (black dots correspond to isolates below); ERCH, error range of clonal heterogeneity (grey region); a.u., arbitrary fluorescence units. Top right, the percentage similarity between basal and induced values for CLASSIC and isolates. Bottom, flow data (green dotted line, basal; green solid, induced), CLASSIC-computed distributions (kernel density (line) calculated from normalized barcode read count (bars)) and part compositions.

To create a library that diversifies the design of this circuit architecture, we identified ten genetic part categories that we hypothesized could have a role in circuit tuning to achieve HFC behaviour. This included four different TA domains^{43–45}, three ZF affinities³⁶ and a set of four intrinsically disordered protein (IDP) domains^{46,47} that have been shown to facilitate liquid–liquid phase condensation of nuclear-localized TFs, and have the potential to drive cooperative, multivalent interactions between synTFs when bound to the promoter^{46,47} (Fig. 2b (left) and Supplementary Fig. 15). We also varied transcriptional regulatory features of the circuit, including four promoters and four terminators in the synTF-coding EU, as well as three core promoters⁴⁸ and different numbers of BMs (2, 4, 8 or 12) in the reporter EU (Fig. 2b (left)). Moreover, we varied the spacing between the EUs (0, 250 or 500 bp) along with their 5′-to-3′ orientation (Fig. 2b (left)) to yield an overall circuit design space of 165,888 compositions (Fig. 2b (right)). This library was constructed in three steps by first assembling protein domain parts to create a level 1 synTF ORF pool, then conducting parallel assemblies to generate level 2 pools of synTF-coding and reporter EUs, and finally combining EU and barcode pools into the level 3 destination vector (Fig. 2b (right) and Extended Data Fig. 1).

Nanopore sequencing of the pooled library yielded barcode assignments for 95.3% of total compositions (Fig. 2c and Extended Data Fig. 2), with a mean of 8.4 barcodes for each circuit (Fig. 2c (bottom left)). We integrated the library into HEK-LP cells and sorted uninduced and 4-OHT-induced populations (8.6 million and 15.7 million cells, respectively) separately into eight bins based on eGFP fluorescence (Fig. 2d (left) and Supplementary Fig. 16). After analysis using Illumina NGS, we matched a total of 121,292 (73% of design space) compositions to barcodes that were detected in the sorted populations (mean barcodes per circuit = 2.25) (Fig. 2c). We then used barcode bin distributions to compute basal, induced and fold change (FC) expression values for each variant. CLASSIC-derived FC values demonstrated excellent overall agreement (MAE = 0.15) (Fig. 2d (top middle)) with those of random isolates ($n = 40$) (Fig. 2d (bottom), Supplementary Fig. 35a and Supplementary Table 2), and values for both basal and induced expression showed comparably high percentage similarities (Fig. 2d (top right)). These results confirm that CLASSIC can generate quantitatively accurate measurements of libraries large enough to profile complex genetic design spaces.

As an incomplete mapping of circuit design space (27% was unmeasured) (Supplementary Fig. 17) could limit the identification of HFC-optimized compositions, we tested whether a complete mapping could be obtained by training an ML model with our data and then using it to predict the behaviour of unmeasured compositions. We trained several model classes (Extended Data Fig. 3) and found that multilayer perceptron (MLP) (90/10 train/validation split, with high-quality test set r^2 of 0.86 for basal and 0.90 for induced data) and random forest (r^2 of 0.84 and 0.86) models significantly out-performed biophysically consistent (r^2 of 0.46 and 0.27) (Supplementary Fig. 18) and unconstrained linear regression (r^2 of 0.53 and 0.64) models (Extended Data Fig. 3a), indicating that the complex, nonlinear interactions between genetic parts can be more accurately captured by advanced model architectures. We selected the MLP to move forward with (Fig. 3a) due to its predictive accuracy and potential for extensibility (Fig. 3a and Methods). We used the MLP to impute basal and induced eGFP expression values to yield a complete design space mapping of all 165,888 compositions (Fig. 3b). We then analysed the distribution of values across a two-dimensional projection of behaviour space (basal versus induced expression), examining the relative densities of compositions in three regions of interest: low basal (<500 arbitrary units (a.u.)), high induced (>70,000 a.u.) and HFC expression (over 25× FC) (Fig. 3b (right)). We observed a greater proportion of library members in the low basal than in the high induced region (around 4.3×), with about 50% of compositions showing FC values of <3×. Only a small fraction of the library (around 8%) fell within the HFC region.

To experimentally assess whether our model can accurately predict behaviour across circuit design space, we constructed cell lines containing individual CLASSIC-measured ($n = 81$) and unmeasured ($n = 15$) compositions, including many predicted to be among the highest FC compositions in the design space (60–80×) (Fig. 3c, Extended Data Fig. 4a, Supplementary Fig. 35b,c and Supplementary Table 2). Flow cytometry analysis of the cell lines closely corresponded with model predictions (MAE = 0.18), confirming the behaviour of numerous HFC compositions predicted to be in the 60–80× range, near the upper boundary of model-inferred behaviour space. Comparison between the MLP-modelled and CLASSIC-measured FC values for the individual variants revealed similar global trends but highlighted the superior accuracy of the MLP model (Extended Data Fig. 4b). Notably, when assessed across the entire behaviour space, we observed a high absolute error (>2) between the model predictions and CLASSIC-measured values at the periphery of behaviour space (Extended Data Fig. 4c), suggesting that the model is capable of correcting measurement errors in the CLASSIC dataset (Extended Data Fig. 4d). These results validate the ability of our CLASSIC-trained ML model to enable gene circuit design space mapping through accurate imputation of inaccurately sampled or unsampled compositions.

With a quantitatively accurate mapping of single-input design space in hand, we sought to uncover part composition rules governing circuit behaviour. First, we examined relative part use frequencies in design space regions of interest (Fig. 3d (top), Supplementary Fig. 19a and Supplementary Table 3). We observed distinct part use in categories associated with the synTF protein (TA domain, IDP domain and ZF affinity), as well as the promoter driving synTF expression, the number of BMs, the core promoter and EU orientation. For example, VPR, a strong TA domain, is used extensively in the high-induced region, but is nearly absent among low basal circuits in favour of a weaker TA domain, p65, while intermediate-strength VP16 and VP64 are enriched among HFC circuits. By the same token, lower- and medium-activity core promoters (miniTK and ybTATA) are excluded from high-activity circuits, while the low basal region has few compositions with mCMV (high-activity core promoter), and HFC circuits use a mixture of ybTATA and mCMV. HFC circuits favoured compositions with the reporter EU in the 5′ position—an orientation that appears to minimize basal circuit expression across design space, potentially by eliminating transcriptional crosstalk from the upstream promoter (Supplementary Fig. 20).

To gain further insights into the design rules underlying HFC behaviour, we used our trained MLP to compute Shapley additive explanation (SHAP) values, a widely used approach that provides an objective metric of each part category's relative importance in determining target behaviour (Fig. 3e). SHAP analysis revealed that HFC behaviour is primarily influenced by part selection within six categories: TA domain, IDP domain, ZF affinity and synTF expression and core promoters, and BM number. To investigate part usage co-variance, we computed cross-category mutual information (MI) for compositions within the HFC region (Fig. 3f and Supplementary Fig. 19b,c). This analysis revealed strong coupling among the six categories, indicating that achieving HFC behaviour requires their coordinated optimization rather than independent tuning. To further explore this interdependence, we applied uniform manifold approximation and projection (UMAP)-assisted k -means clustering on HFC compositions, identifying three clusters (Fig. 3g, Methods and Extended Data Fig. 5a–c). Notably, all three clusters shared a common locus design that maximizes the FC between the induced and basal states. This included features that enhance circuit output, such as high BM valency ($n = 8$ or 12), a high-activity core promoter (mCMV) and a preference for the T7 terminator, as well as features that reduce basal expression, such as reporter-to-coding EU orientation, and extended spacing (500 bp) between EUs. The largest cluster (A), which achieved the highest median FC (38.4-fold; Extended Data Fig. 5d), paired medium-activity promoters driving synTF expression (hEF1a1 and hPGK) with medium-affinity ZFs and medium-strength

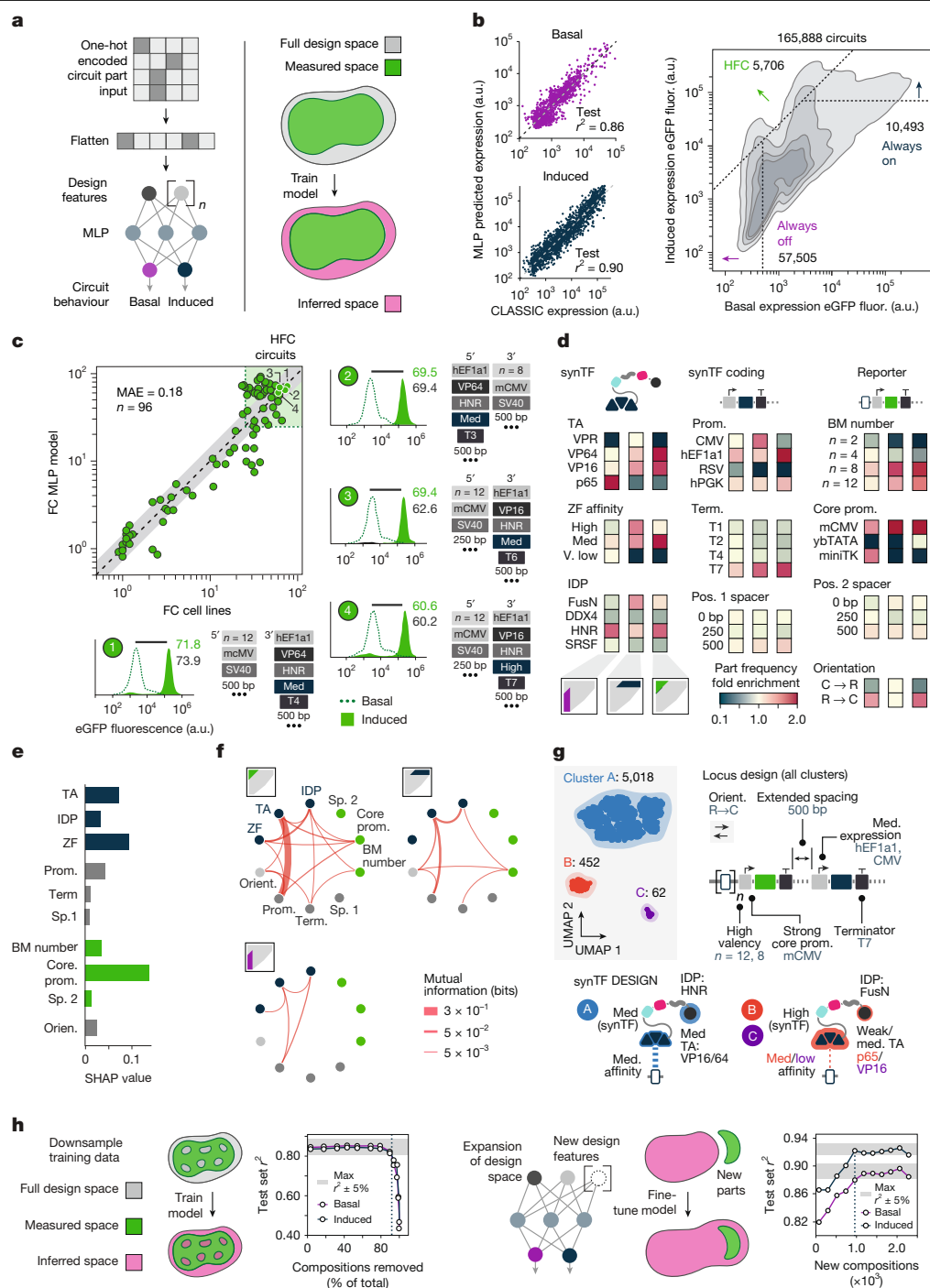


Fig. 3 | ML-aided mapping of single-input circuit design space reveals gene circuit design rules. **a**, The workflow and ML framework for ML-guided design space mapping. Left, data encoding and training of a MLP modelling inducible synTF circuit behaviour space. Right, data from 121,000 measured compositions (green) within -166,000-member design space (grey) were used to train the MLP to re-fit circuit behaviour for measured compositions and infer unmeasured space (pink). **b**, ML model of single-input design space. Left, MLP predictions versus basal (purple) and induced (navy) CLASSIC measurements (Pearson's r^2). Right, contour plot of MLP-modelled basal and induced expression (contours, light to dark: 97.5%, 90%, 70%, 50% of measurements). Dotted regions: low basal (<500 a.u.; purple arrow); high induction (>70,000 a.u.; navy arrow); HFC (>25; green arrow). The numbers denote the circuit counts per region. Fluor., fluorescence. **c**, Experimental validation. Top left, the FC values for cell lines from CLASSIC-measured configurations versus model predictions ($n=1$ for all data points in scatter). The white outlines highlight individual cell lines shown below; the grey region shows ERCH; the green box shows the HFC behaviour

region. Right and bottom left, flow cytometry analysis of cell lines (green dotted line, basal; solid green, induced; green text, FC) with ML-predicted FC (horizontal bar; black text, FC) and circuit composition. **d**, Single-input genetic part use across low basal, high induced and HFC regions (left to right). Fold enrichment was observed over the expected part occurrence from a balanced library. Prom., promoter; pos., position. **e**, Feature importance. Part-category-specific absolute SHAP values for HFC compositions. Larger SHAP values indicate stronger contributions to model predictions of HFC behaviour. Sp. 1, synTF spacer; Sp. 2, reporter spacer. **f**, The MI between part categories across behaviour regions. The line thickness denotes the MI magnitude. Orient., orientation. **g**, HFC circuit clustering. Top left, UMAP projection of HFC designs. Right, Mechanistic summary of HFC design strategy. **h**, Model efficiency and adaptability. The model performance on basal (purple) and induced (navy) test-set measurements (Pearson's r^2) versus the training data fraction for existing (left) and new parts (right; NFZ data) is shown. The dark grey box shows the maximum $r^2 \pm 5\%$; the dotted line shows the 95% maximum r^2 threshold.

TA domains (VP64 and VP16) containing the HNR IDP domain. Performing a dose–response analysis on a representative composition from this cluster revealed a Hill coefficient (n_H) of 2.13 (Supplementary Fig. 21), indicating moderate cooperativity, probably due to HNR-mediated interactions between synTFs⁴⁹. By contrast, compositions in clusters B and C favoured higher synTF expression (CMV promoter), with B containing medium-affinity ZFs and a weak TA domain (p65), and C showing enrichment for low-affinity ZFs and a medium-strength TA domain (VP16). Taken together, CLASSIC-enabled design space mapping indicates that HFC behaviour is enabled by specific locus designs coupled with synTF architectures that balance expression level, ZF affinity, transcriptional activation and cooperative binding effects that minimize basal and maximize induced circuit activity. These analyses further reveal that HFC does not arise from a single optimal solution, but from multiple balanced combinations of parts that achieve similar phenotypic outcomes through subtly different molecular strategies, with medium-activity components providing the most reliable overall route to circuit performance.

Although our results establish that accurate design space mappings can be obtained for regimes with balanced training-to-prediction ratios, we realized that models trained with CLASSIC data would offer greater use if they could generalize to a broader range of out-of-sample test cases. To evaluate the model's generalization capacity, we progressively downsampled the training data while performing validation with the held-out test set (Fig. 3h (left)). We found that only around 9% of the data were sufficient to achieve an r^2 that was 95% of the maximum (>0.81). We then examined whether the design space mapping could be expanded by incorporating additional genetic part measurement data (Fig. 3h (right) and Extended Data Fig. 6a). To do this, we assembled two libraries, each containing approximately 3,500 members: one incorporating a previously unmeasured TA domain (NFZ)⁵⁰ (Extended Data Fig. 6b) and another with a 10-GS linker replacing the IDP domain (Extended Data Fig. 6c). After measurement of these libraries (Extended Data Fig. 6d–f), we fine-tuned the existing base model with the new data to predict the expanded design space, which now included the new parts (approximately 259,000 compositions) (Extended Data Fig. 6g,h). Whereas the base model alone showed good predictions of new part behaviour (r^2 of around 0.82 for basal and 0.87 for induced conditions for NFZ-containing compositions), the addition of the new data improved model generalization, achieving 95% of maximum accuracy (r^2 of around 0.88 for basal and 0.92 for induced conditions) with just around 1,000 instances (Extended Data Fig. 6g,h). These results show that our modelling framework is highly data efficient and can easily and rapidly incorporate new parts and design features through fine-tuning. This suggests its potential, through multiple cycles of library construction and testing, to generalize to design spaces that are much larger than the datasets used for training (Extended Data Fig. 6i and Supplementary Fig. 35d).

We attempted to evaluate these capabilities for design space expansion by mapping a design space for multi-input circuits with two inducible synTFs—one regulated by 4-OHT and the other by grazoprevir (GZV), a small-molecule protease inhibitor that blocks autolytic cleavage of a NS3 protease-synTF fusion⁵¹ (Fig. 4a and Supplementary Fig. 22). In this 3-EU architecture, co-regulation of the reporter locus by the two synTFs could potentially support Boolean logic, including AND-gate function if the synTFs cooperatively bind to and activate the reporter, and OR-gate function if they independently maximize activation. We hypothesized that this architecture could be tuned to achieve these behaviours by diversifying a set of 68 design features, including many that were chosen based on our results from the single-input library, as well as synTF BMs arranged in four distinct patterns, yielding in an overall design space of approximately 3.4 billion compositions (Fig. 4b). To create a mapping for this space, we used a two-step data collection strategy (Supplementary Fig. 23), first measuring a densely sampled base library space comprising a 44-feature subset (1.1 million

compositions), and then obtaining data for an expansion library that sparsely sampled the remaining features (Fig. 4b (bottom)). During base library construction, we deliberately constrained diversity to ensure a high read depth (Supplementary Fig. 24), obtaining barcode indexes for around 9% (approximately 112,000 compositions) of the 1.1 million potential compositions (Fig. 4c and Supplementary Fig. 25), with a mean of 3.2 barcodes for each circuit (Fig. 4c (bottom left)). The expansion library was constructed by assembling 24 independent 1,500-member pools, each containing one of the expansion features held constant against a random sampling of all other base library features (Supplementary Fig. 26). Both libraries were integrated separately into HEK-LP cells and left untreated or treated with 4-OHT, GZV or both inducers (Supplementary Figs. 27 and 28). Each treatment was sorted into eight bins based on eGFP fluorescence and isolates were obtained from the base ($n = 14$) and expansion ($n = 13$) libraries. As before, we performed Illumina NGS to obtain expression values from both libraries (around 83,000 and 45,000, respectively, for base and expansion libraries) (Fig. 4c (bottom)), once again observing overall agreement between CLASSIC and isolate measurements (MAE = 0.28, 0.23) (Fig. 4d and Supplementary Figs. 29 and 30).

To create a multi-input circuit design space mapping, we first trained a five-layer MLP (Fig. 5a) with the base library data, yielding a model with strong predictive accuracy (90:10 train/validation split with test set r^2 values of 0.80, 0.90, 0.86 and 0.88 for basal, 4-OHT, GZV and both inducer predictions) (Extended Data Fig. 7). We then extended the model to the full 3.4×10^9 -member design space by fine-tuning with expansion library data, resulting in good predictive performance (r^2 values of 0.79, 0.84, 0.84 and 0.82) (Fig. 5b (left) and Supplementary Fig. 31). In accordance with other recent studies^{52,53} we found that our two-step strategy of fine-tuning on a pretrained base model demonstrated superior performance compared with training from scratch on all data (Supplementary Fig. 31c). We calculated the similarity of circuit composition behaviour across the design space to that of idealized AND- and OR-gates using Kullback–Leibler divergence (Methods). We found that, while only a small region of base library space demonstrated Boolean-like behaviour (D_{KL} values of <0.12 for AND, <0.05 for OR), these regions expanded in the complete design space (Fig. 5b (right)). To validate the model predictions, we constructed 36 individual compositions spanning the behaviour space, 35 of which were unmeasured in our CLASSIC dataset, including many predicted to exhibit AND- and OR-like function (Fig. 5c, Extended Data Fig. 8, Supplementary Fig. 37 and Supplementary Table 2). As before, we observed good agreement between measurements of individual compositions and the model (MAE = 0.32) (Fig. 5c), confirming the effectiveness of our data sampling and training approaches.

Similar to the single-input design space, we observed strong polarization in part use within both AND- and OR-like regions (Fig. 5d). While compositions from both regions were enriched for a high number of BMs ($n = 8$ or 12) and strong core promoters (mCMV or SCP2), AND-like compositions predominantly used clustered BM patterns, whereas OR gates used exclusively interspersed patterns. In both regions, the 4-OHT-inducible synTF (synTF A) was expressed at intermediate levels (hEF1a1) while the GZV-inducible synTF (synTF B) showed stronger expression and greater overall part diversity. SHAP analysis revealed that core promoter and inter-EU spacing were common drivers of both behaviours (Fig. 5e), whereas OR-like behaviour was impacted by more features than AND-like behaviour, indicating a greater degree of feature interactivity. This was consistent with part use coupling analysis, which showed a lower degree of interdependence among AND-like compositions, with strong coupling between BM pattern, both synTF expression promoters and synTF A affinity (Fig. 5f (left)). The higher intercategory coupling in the OR-like region was driven by strong interactions between features on different synTFs (promoters and TA domains) (Fig. 5f (right)), suggesting that co-optimization of expression level and activation strength is important for tuning OR-like circuit behaviour.

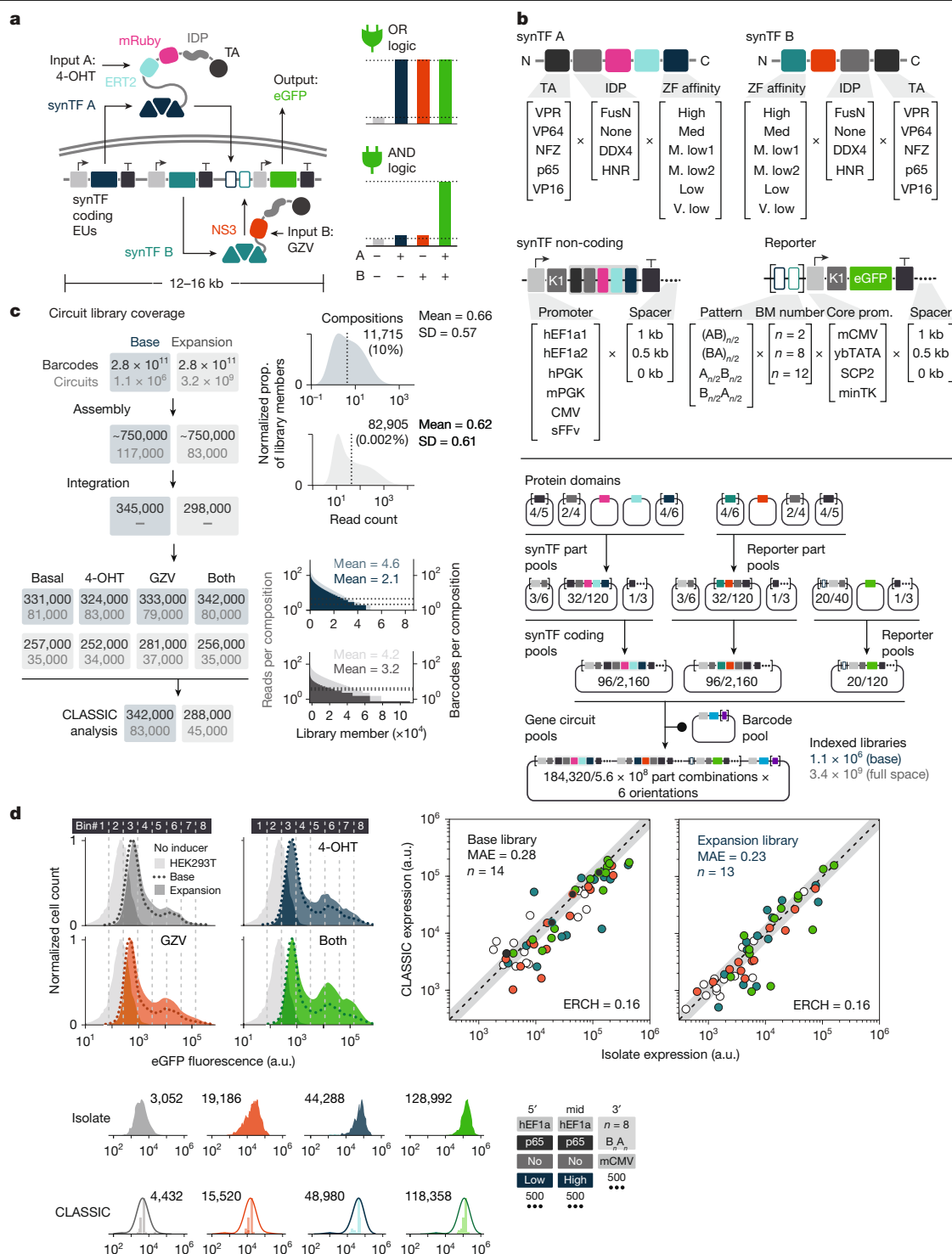


Fig. 4 | ML-guided exploration of multi-input gene circuit behaviour.

a, Circuit diagram of the multi-input inducible synTF. Left, the circuit contains a 4-OHT inducible synTF A (navy), a GZV-inducible synTF B (teal) and an eGFP reporter unit (green). 4-OHT (input A) binds to the ERT2 domain (light blue), triggering nuclear translocation of synTF A and eGFP activation (output). GZV (input B) inhibits NS3 protease activity (orange), preventing synTF B cleavage and allowing eGFP activation (output). Right, OR (top) and AND (bottom) Boolean operations. **b**, Multi-input circuit design space. Top, synTF components; middle left, non-coding parts; middle right, reporter parts. ZF affinity mutants: high, wild type; med, ΔC ; med low1, WT 1 $\times$; med low2, WT 3 $\times$; low, $\Delta C2$; very low, ΔC 6 $\times$. Bottom, assembly of full-length circuits from protein domains; numbers within plasmids denote variant counts in base/full design spaces. **c**, Library coverage, balance and indexing. Left, circuit (light grey text, bottom) and barcode (dark grey text, top) recovery through CLASSIC workflow for base (light blue boxes) and

expansion (light grey boxes) libraries. Top right, Nanopore sequencing reveals 10% and 0.002% coverage of the base (light blue histogram) and expansion (light grey histogram) design spaces, respectively. Bottom right, Nanopore data ranked by reads per composition (base: light blue, expansion: light grey) and barcodes per composition (dark blue, base; dark grey, expansion). Numbers are rounded to the nearest thousandth. **d**, Library sorting and measurement. Top left, eGFP histograms for base (dotted) and expansion (shaded) libraries under no inducer (grey, top left), 4-OHT (navy, top right), GZV (orange, bottom left) or both (green, bottom right). Sorting binds are shown by vertical grey dotted lines; HEK293T cells are represented by the light grey histogram. Top right, CLASSIC versus ground-truth eGFP expression for isolates from the base ($n = 14$, left) and expansion ($n = 13$, right) libraries under all four conditions ($n = 1$ for all data points in scatter). Bottom, flow cytometry histograms (top), CLASSIC-computed distributions (bottom) and composition ID for one base-library variant (grey fill in scatter).

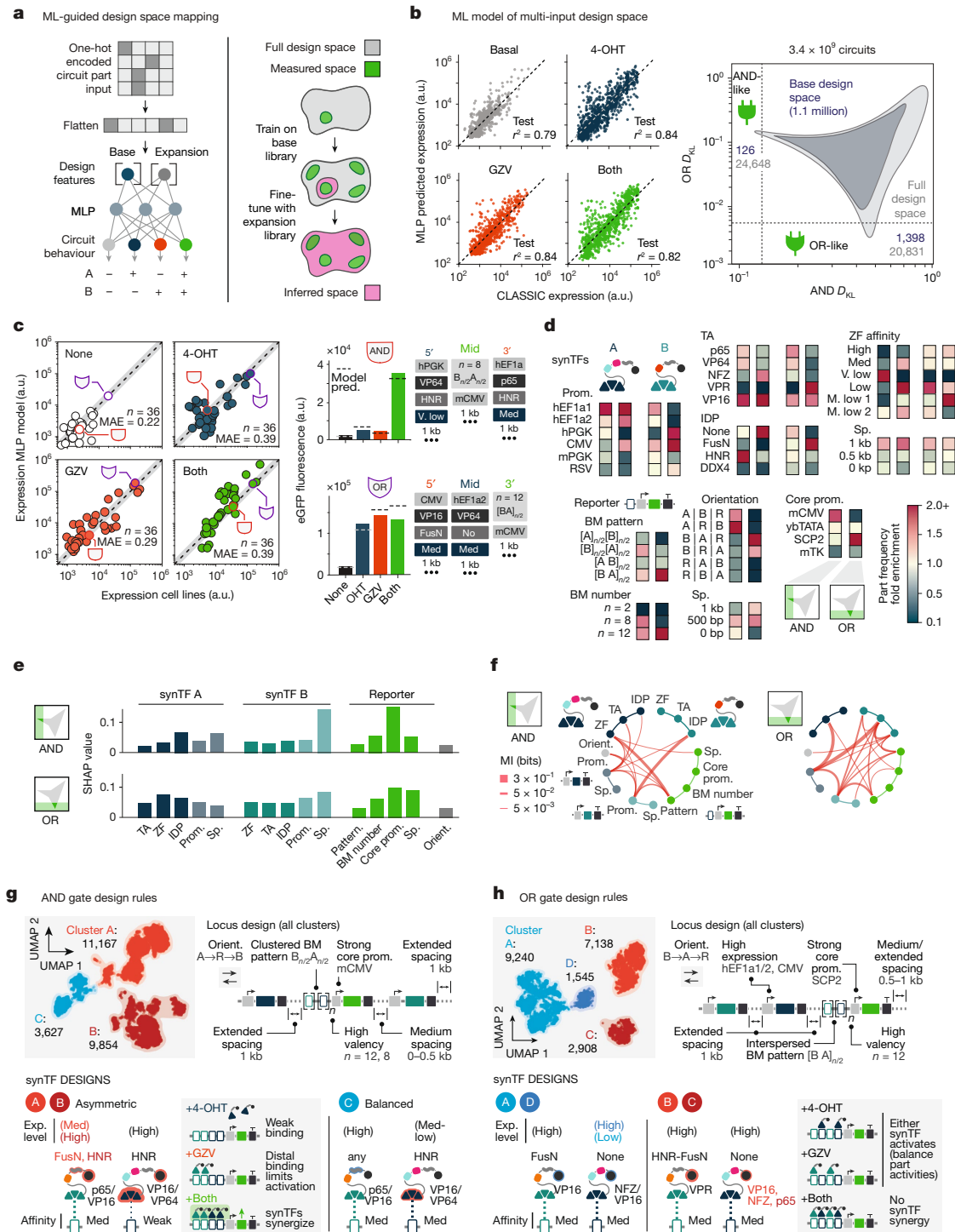


Fig. 5 | Analysis of digital logic gene circuit design rules in $>10^8$ -member design space. **a**, Design space inference. Left, workflow for predicting multi-input circuit behaviours (light grey, basal; navy, 4-OHT; orange, GZV; green, both). Right, training on a base library (green, top) enables prediction (pred.) of a subset (pink, middle) of the full space (light grey). Additional data (green, middle) enable inference of the entire space (pink, bottom). **b**, Model validation. Left, test set r^2 for the fine-tuned model under four input conditions (colours are as defined in **a**). Right, compositions from the base (dark grey) and full (light grey) design spaces, plotted as Kullback–Leibler divergences (D_{KL}) from archetypal AND and OR behaviours. Compositions below or left of the dotted lines represent extremely OR- and AND-like designs, respectively. **c**, Experimental validation. Top, the predicted versus measured eGFP expression for 36 individually constructed compositions (white, basal; navy, 4-OHT; orange, GZV; green, both) (MAE = 0.32) (top) ($n = 1$ for all data points in scatter). The most AND-like

(red) and OR-like (purple) designs are outlined (right). **d**, Two-input circuit part use. Genetic part use in AND-like (left) and OR-like (right) regions, grouped by synTF (A (left) and B (right)) and calculated as in Fig. 3d. **e**, Feature importance for part categories in AND- (top) or OR-like (bottom) regions. Larger values indicate a greater contribution to predictions of logical behaviour (navy, synTF A; dark grey, synTF coding; teal, synTF B; light blue, synTF B coding; green, reporter). **f**, Part category MI in AND-like (left) and OR-like (right) regions, indicated by the thickness of the red line (colours are as defined in **e**). **g**, Clustering AND-like circuits. Top left, UMAP-projection. Top right, common locus design (navy, synTF A; green, reporter; teal, synTF B). Bottom, cluster A (light red), B (dark red) and C (blue) design strategy overviews. Exp., expression. **h**, Clustering OR-like circuits. Top left, UMAP projection. Top right, common locus design (colour is as defined in **g**; right). Bottom, cluster A/D (blue) and B/C (red) design strategy overviews.

Clustering analysis of AND-like compositions yielded three major clusters. As with HFC circuits from the single-input design space, locus design remained constant across clusters (Fig. 5g, Methods and Extended Data Fig. 9a–c). The largest subset of AND-like compositions (clusters A and B) featured a low-affinity synTF A binding to BMs near the promoter, while synTF B, which contained a medium-affinity ZF and a weak TA domain, bound distally. This suggests a mechanism in which synTF B, separated from the core promoter by around 110 bp, is unable to independently activate transcription but can facilitate activation by reinforcing weakly-binding synTF A, potentially through cooperative IDP-mediated interactions between the two synTFs (HNR on synTF A, and FusN on synTF B (cluster A) or HNR on both (cluster B)) (Fig. 5g (right)). While this ‘asymmetric’ activation mechanism comprises the majority of AND-like compositions, we observed a third, minority solution family (14%) that uses more balanced expression levels and affinities of IDP-domain-containing synTFs to achieve AND-like behaviour, and may depend on fine-tuning of component interactions, and is therefore less robust to part substitutions (Extended Data Fig. 9b). Despite its lower representation in AND-like behaviour space and lower average performance, we were able to validate that one of our constructed cell-lines from this cluster displayed AND-like behaviour (Extended Data Fig. 8 (composition 19)). Clustering of OR-like compositions revealed four clusters, all with the same locus design (distinct from the AND-gate design), that activate to intermediate levels, either individually (–/+ , +/– induction conditions) or in combination (+/+) (Fig. 5h and Extended Data Fig. 10a–c). This is probably facilitated by interspersed BM binding arrays, which provide both synTFs with similar access to the core promoter (Fig. 5h and Extended Data Fig. 10a). Medium-affinity synTF binding is a common feature of all four clusters, and a lack of IDP domains for synTF A suggests that either synTF–synTF interactions have a limited role in reporter activation, or that IDP-domain-mediated interactions between synTF B serve to occlude synTF A from the locus under +/+ induction conditions, thereby impeding additive activation. In summary, our mapping of multi-input circuit design space enabled the identification of non-intuitive compositional rules, and offered insights into the role that different genetic parts play in determining circuit function.

Here we established the utility of combining long- and short-read sequencing to perform massively parallel quantitative profiling of multi-kb genetic part compositions in human cells. CLASSIC has potential for exploring the emergence of phenotypic function from genetic module composition across a range of organizational scales and phylogenetic contexts, including viruses⁵⁴, bacterial operons⁵⁵ and chromatin domains^{55,56}. As it uses broadly accessible molecular cloning and experimental analysis pipelines, CLASSIC can be readily adopted by investigators working on diverse phenotypic screening tasks (Supplementary Fig. 32). By enabling high-throughput profiling of diverse genetic part compositions, CLASSIC expands the scope of inquiry for synthetic biology projects, reducing the number of DBTL cycles and cost required to identify optimized part compositions across medium- to large-scale design spaces. CLASSIC is also well suited for underdetermined systems, in which iterative design space expansion can enable convergence to difficult-to-access target behaviours. We further show that CLASSIC offers an affordable way to test vastly more genetic designs in a single experiment, although traditional approaches may be more cost-effective for smaller-scale projects (<100 constructs) (Supplementary Fig. 33). Given the flexibility and scalability of this platform, we anticipate that it will accelerate genetic design for synthetic biology applications across a spectrum of organismal hosts that require probing complex design spaces to achieve optimal behaviour, from the development of bioproduction strains⁵⁷ to multigenic cell therapy programs¹².

Finally, we showed that CLASSIC-derived ML models can accurately predict out-of-sample and edge-case circuit behaviours and, with appropriate training strategies, generalize to orders-of-magnitude

larger design spaces. This demonstrates the utility of using a data-driven approach to solve complex circuit design challenges and identify designs that support target circuit behaviour (Supplementary Figs. 34–37), while revealing underlying, non-intuitive design principles that are challenging to capture using biophysical models. While previous work has used ML approaches to develop sequence-to-function models for individual genetic parts^{3,58,59}, our study lays the groundwork for developing artificial-intelligence-based models of gene circuit function that treat part compositions as features. In the future, integration of CLASSIC data with sequence-to-function and genomics datasets could support training high-capacity deep-learning architectures that operate over large genetic length scales (for example, transformers, graph neural networks or diffusion models) for context-aware generative circuit design, either in a black-box manner, without the incorporation of regulatory or biophysical priors, or synergistically with existing analytical frameworks to create interpretable models that provide mechanistic insights into genetic design⁶⁰.

Online content

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Methods

Plasmid library construction

Circuit libraries were cloned using a custom hierarchical Golden Gate³³ assembly scheme, which enables rapid, modular cloning of complex gene circuits starting from individual genetic part sequences (Supplementary Figs. 2–4 and Supplementary Table 1). In brief, input DNA fragments are amplified from genomic or commercial DNA sources and cloned into kanamycin-resistant sub-part entry vectors using Bpil (Thermo Fisher Scientific) (level 0). Sequence-verified sub-part plasmids are then used as inputs for assembly into carbenicillin-resistant (AmpR) entry vectors using BsaI-v2 HF (NEB) to yield genetic part plasmids (that is, promoters, ORFs, terminators) (level 1). EUs are then constructed by assembling promoter, ORF and terminator part plasmids into a kanamycin-resistant entry vector using Esp3I (Thermo Fisher Scientific) (level 2). EUs are subsequently combined into multi-unit arrays by assembling into a carbenicillin- or spectinomycin-resistant (SpectR) destination vector using Bpil (Thermo Fisher Scientific) (level 3).

In this system, barcode pools are incorporated into library assemblies at level 2 and 3 (Supplementary Fig. 5). Level 2 barcoded EU pools were generated by first constructing a destination vector carrying a ccdB placeholder expression cassette downstream of the BFP stop codon. A semi-degenerate 18 bp barcode oligo pool (IDT)⁶¹ was polymerase-extended and cloned in place of the ccdB cassette using BsaI-v2 (NEB) in a 20 µl Golden Gate reaction: 2 min at 37 °C and 5 min at 16 °C for a number of cycles equal to 10× the number of input fragments, followed by a 30 min digestion step at 37 °C and subsequent sequential 15 min denaturation steps at 65 °C and 80 °C. The resulting assemblies were purified using a miniprep column (Epoch Life Sciences), electroporated (Bio-Rad) into 100 µl NEB 10-beta electrocompetent *Escherichia coli* (NEB), and plated onto a custom 30 in × 24 in LB agar + kanamycin (LB_{Kan}) plate, yielding around 120 million colonies. The plates were grown at 37 °C for 16 h, at which time the colonies were scraped into LB_{Kan} (20 ml), incubated for 30 min at 37 °C and the plasmid library was extracted by miniprep (Qiagen). For large-scale libraries, level 3 barcode pools were generated to ensure a unique barcode-to-circuit mapping. To create level 3 barcode pools, a semi-degenerate 18 bp oligo pool (IDT) was polymerase-extended and cloned by Golden Gate assembly into a SpectR destination vector carrying a placeholder ccdB cassette using PaeCI (NEB) according to the assembly protocol above. The resulting assemblies were transformed into 100 µl in-house generated chemically competent^{62,63} StbI3 *E. coli* cells and plated onto a 10 cm LB agar + spectinomycin (LB_{Spect}) plate. Plates were grown at 37 °C for 20 h to yield approximately 15,000 colonies, followed by colony scraping and plasmid DNA extraction, as described above.

Part pools (levels 1), EU pools (level 2) and circuit libraries (level 3) were constructed by combining input plasmids at 50 fmol for each part category (15 µl total volume) using the above-described cycling protocol. Exact sample sizes for each library were not predetermined. We aimed to get between 10× and 100× coverage of our library at each of the stages in the CLASSIC process: construction, genomic integration, and phenotypic selection. Thus, transformations varied by library size: level 2 and 3 assemblies for the EU library (Supplementary Fig. 9b,c) were respectively transformed into 100 µl or 200 µl of chemically competent *E. coli* (DH5a and StbI3) and plated onto LB_{Kan} or LB agar + carbenicillin (LB_{Carb}) plates for 12–16 h (37 °C) to yield around 15,000 colonies, respectively. Colonies were scraped into 13 ml LB_{Kan} or LB_{Carb} and miniprep using the Qiagen kit. Level 1 and 2 assemblies for the single-input library (Fig. 2b and Extended Data Fig. 1a,b) were transformed using 100 µl and 400 µl of chemically competent *E. coli* (StbI3) and grown on LB_{Carb} and LB_{Kan} agar plates at 37 °C for 12–16 h to yield around 5,000 and 20,000 colonies, respectively. Level 3 single-input base library assemblies were purified using a miniprep column (Epoch Life Sciences), electroporated (Bio-Rad) into 100 µl 10-beta electrocompetent cells (NEB), and then plated onto ten 15 cm LB_{Spect} agar

plates for 16–20 h (37 °C) to yield around 3 million colonies. Each level 3 single-input fine-tuning library assembly was transformed into 500 µl of chemically competent *E. coli* (StbI3) and grown on LB_{Spect} agar plates at 37 °C for 12–16 h to yield around 40,000 colonies (Extended Data Fig. 6b,c).

Level 1 assemblies for the multi-input base library (Fig. 4b and Supplementary Fig. 24a,b) were treated identically to those from the single-input library. All level 2 assemblies associated with the base design space were transformed into 100 µl of chemically competent *E. coli* (DH5a) and grown on LB_{Kan} agar plates at 37 °C for 12–16 h to yield around 200 colonies. Then, 20 (reporter EUs) or 32 (synTF A or BEUs) colonies from each level 2 library were picked separately into 1 ml of TB_{Kan} and shaken overnight at 37 °C and 900 rpm. Next, 200 µl from the 20 or 32 cultures associated with each level 2 library was combined and miniprep using a Qiagen kit. Separate level 3 base library assemblies were carried out for each of the six possible circuit orientations, and purified using a miniprep column (Epoch Life Sciences), electroporated (Bio-Rad) into 15 µl of electrocompetent cells (10-beta, NEB) and then plated onto 15 cm LB_{Spect} agar plates for 16–20 h (37 °C) to yield around 200,000 colonies. Cells were scraped and miniprep separately using the Qiagen kit.

Level 1 assemblies for the multi-input expansion library were generated separately for each new part (24 in total) by combining balanced pools of level 0 part plasmids while holding the new parts constant (Fig. 4b and Supplementary Fig. 27a,b). After transforming, scraping and miniprep these new part libraries as described above, level 2 assemblies were generated for each new part, transformed into 20 µl of chemically competent *E. coli* (DH5a) and grown on LB_{Kan} agar plates at 37 °C for 12–16 h to yield around 50–100 colonies. To sample compositional diversity within each new part library, 10 colonies from each plate were picked into 1 ml of TB_{Kan}, shaken overnight at 37 °C and 900 rpm, pooled and miniprep using the Star Liquid Handler (Hamilton) to generate 72 level 2 new part libraries. Each of these libraries was then assembled with five-member level 2 pools from the base library that corresponded to positions not containing the EUs with the new parts, along with the BFP barcode EU pool. These level 3 assemblies were then pooled on the basis of their orientation, column purified, electroporated (Bio-Rad) into 7.5 µl of 10-beta electrocompetent cells (NEB), and then plated onto 10 cm LB_{Spect} agar plates for 16–20 h (37 °C) to yield around 100,000 colonies. The libraries were scraped and miniprep separately using the Qiagen kit.

Long-read plasmid sequencing

To generate indices linking assembled constructs to their associated DNA barcodes, we used Oxford Nanopore Technology (ONT) long-read sequencing^{64,65}. Libraries were prepared for long-read sequencing by digesting 2 µg of a level 3 plasmid library using Esp3I (Thermo Fisher Scientific) and purifying linearized fragments using a 0.5× volume of magnetic beads (Omega Bio-Tek) by volume. Nanopore sequencing adapters were added to the linearized DNA pool using the LSK-112 or LSK-114 ligation sequencing kit (ONT) and sequenced using on a MinION Mk1b device (ONT) equipped with an R10 flow cell (FLO-MIN112) or R10.4.1 flow cell (FLO-MIN114). Basecalling was performed using Guppy or Dorado (ONT, super high accuracy mode) running on a GPU (Nvidia RTX 3090). Composition and barcode assignment was performed using WIMPY (what's in my pot, y'all) (Supplementary Fig. 6), a custom MATLAB analysis pipeline composed of six discrete modules that converts basecalled Fastq files into a part composition array and barcode sequence for each Nanopore read (Supplementary Fig. 6). In brief, WIMPY first uses Fastqall (1) to import and consolidate basecalled fastq files, followed by Bowties (2) to index the reads to a constant region in the level 3 plasmid backbone. In this study, the puromycin-resistance cassette (PuroR) was designated as the constant region and, after indexing to PuroR, reads were filtered to remove incomplete assemblies and non-library fragments. WIMPY then calls Tilepin (3), which uses

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localized containment searching (also known as tiling) to identify important genetic landmarks, such as ORFs, to define the different sections of the sequence (for example, reporter array, synthetic TF coding region or barcoded BFP EU). Chopat (4) then truncates the reads into smaller arrays based on the previously identified landmarks to aid later identification of individual parts. WIMPY then calls Viscount (5a) or FASTar (5b). Viscount determines compositions for filtered reads by identifying and assigning genetic parts through a combination of general Smith–Waterman alignments⁶⁶ and localized containment searches. This is done by splitting the reference sequence into 6–10 bp tiles with a 1 bp stride, after which Nanopore reads are queried for the number of tiles contained for each part reference sequence. FASTar uses the same localized containment searches but indexes the location and number of observed reference sequences, and is used to identify repeated sequences, such as the number of BMs in the single-input library or BM patterns in the dual-input library. Reads containing >3% of tiles for a reference sequence are assigned while reads with more than one reference assignment are discarded. Barcode sequences for each read are concurrently determined by Barcoat (5c), in which the sequence section downstream of the BFP EU is aligned to a degenerate reference sequence using a custom alignment matrix.

Cell culture

HEK293T cells (ATCC CRL-11268) used in this study were cultured under humidity control at 37 °C with 5% CO₂ in medium containing Dulbecco's modified Eagle medium (DMEM) with high glucose (Gibco, 12100061) supplemented with 10% FBS (GeminiBio, 900-108), 50 U ml⁻¹ penicillin, 50 µg ml⁻¹ streptomycin (Gibco, 15070063), 2 mM L-alanyl-L-glutamine (Caisson Labs, GLL02), hereafter, complete DMEM. HEK293T-LP cells without and with integrated libraries were maintained in complete DMEM supplemented with 50 µg ml⁻¹ hygromycin B (Sigma-Aldrich, H3274) and 1 µg ml⁻¹ puromycin (Sigma-Aldrich, P8833), respectively. All cell lines tested negative for *Mycoplasma*.

Single-copy library integration

To establish a landing pad (LP) cell line, low-passage HEK293T cells were co-transfected with a dual Cas9 and gRNA expression vector (pX330-T2) targeting the human *AAVS1* locus⁶⁷ and a PmeI-linearized (NEB) repair template comprising a constitutively expressed YFP-HygR-expression cassette with a BxB1 *attP* recognition site flanked by 800 bp homology arms for the human *AAVS1* locus (pROC079) (Supplementary Fig. 7a). Genomic integration events were selected using 50 µg ml⁻¹ hygromycin B, and YFP⁺ cells were flow sorted to isolate clones (WOLF, NanoCellect). Clones were expanded to 24-well plates and tested for integration competency by integration of an *attB*-containing destination vector and subsequent flow cytometry, as well as for the presence of the intact LP cassette at the *AAVS1* locus using PCR (Supplementary Fig. 7c and Supplementary Table 1). gDNA was extracted from each clonal population (Omega Bio-Tek) and 50 ng of gDNA was used as the template in a 25 µl reaction along with 0.25 µl of Q5 polymerase (NEB), 0.5 µl of 10 mM dNTPs, 5 µl of 5× Q5 polymerase buffer and 1 µl of each 10 µM genotyping primer. Primers were specific to the native *AAVS1* locus, LP or destination vector. The thermocycling protocol was 2 min at 98 °C, followed by 30 cycles of 98 °C for 20 s, 65 °C for 20 s and 72 °C for 1 min, with a final step of 72 °C for 5 min. A list of all of the genotyping primer sequences is provided in Supplementary Table 1. We identified a clone (LP79 36) that demonstrated BxB1-dependent, single-copy integration of vectors containing an *attB* recognition site and used this cell line for all of the following experiments. We verified single-copy genomic integration of the LP cassette by droplet digital PCR using primer sets for the engineered locus (Hygro) and a pair of previously validated native genes (beta actin and *POTP*) (Supplementary Fig. 7d). In brief, each 25 µl reaction was composed of 12.5 µl of EvaGreen Supermix (Bio-Rad), 1.25 µl of each primer (diluted to 2 µM), 20 ng of gDNA and 0.1 µl of HindIII (NEB). Droplets were generated in the QX600 Droplet

Generator (Bio-Rad) by combining 20 µl of each reaction with 70 µl of Droplet Generation Oil (Bio-Rad). PCR was performed on 40 µl of the emulsified reaction using the following protocol: 5 min at 95 °C, then 40 cycles of 95 °C for 30 s and 60 °C for 1 min, followed by 5 min at 4 °C and 5 min at 90 °C. The PCR results were then analysed using the QX600 Droplet Reader (Bio-Rad). All of the resulting concentrations were normalized to the HEK293T beta actin control.

To integrate individual constructs, 1.25×10^5 cells were plated into a 24-well plate 1 day before transfection. The well was co-transfected with 250 ng of a level 3 vector containing a construct of interest and 125 ng of BxB1 expression plasmid (Addgene, 51271) using JetPrime (VWR) (Supplementary Fig. 7b). pCAG-NLS-HA-Bxb1 was a gift from P. Pelczar (Addgene, 51271)⁶⁸. Then, 2 days after transfection, cells were passaged into complete DMEM containing 1 µg ml⁻¹ puromycin and selected for 10 days to achieve homogenous expression. Library integration was performed by co-transfecting HEK293T-LP cells with 250 ng of plasmid library and 125 ng of BxB1 per 200,000 cells (384-member library = 1 million cells transfected; 166,000-member base library = 100 million cells transfected; 166,000-member sublibrary = 10 million cells transfected; 3.4×10^9 -member base library = 200 million cells transfected; 3.4×10^9 -member expansion library = 200 million cells transfected) using JetPrime. Medium exchange was performed after 8 h and cells were expanded for an additional 40 h, split into five culture dishes and grown for 6–8 days under puromycin selection. Cells were passaged 1:3 and cultured for an additional 5 days under puromycin selection, and then combined in complete DMEM for flow sorting (10 million cells per ml).

To determine the clonal variability within LP-integrated cell populations, we integrated a constitutively expressed mCherry EU according to the protocol described above. We then randomly sorted 23 clones from the population (WOLF, NanoCellect) and measured their mCherry expression (Supplementary Fig. 7e). We calculated the MAE for this set (MAE = 0.08) and used this value to define an expected error range from clonal heterogeneity (ERCH), which represents the expected variability of cells from an integrated population. For the 384-member library, the expression ERCH was set to 0.08 on either side of the expected value. For all libraries constructed as part of the single-inducer circuit class, the FC ERCH was calculated to be 0.16 on either side of the expected value. For libraries associated with the multi-input design space, the expression level ERCH was calculated to be 0.16 on either side of the expected value.

Flow sorting

To prepare libraries for flow sorting, cells were lifted using TrypLE (Gibco), washed twice with PBS and resuspended in complete DMEM (10 M cells per ml). For the 384-member library, the mRuby expression distribution was divided into 10 evenly log-spaced bins on the Sony MA900 system set to purity mode (Supplementary Fig. 11). The number of cells collected for each bin was proportional to the percentage of the mRuby distribution expressed in that bin, with a target of around 125,000 cells sorted in the most populous bin (for example, 2,252 cells were collected for bin 1 (0.53% of library), 126,562 for bin 4 (29.78% of library)). For the single-input base and fine-tuning libraries, cells were split and grown in DMEM either with 1 µM 4-OHT (Sigma Aldrich) or without 4-OHT for 72 h (Supplementary Fig. 16 and Extended Data Fig. 6e). The eGFP expression distributions for both conditions were then divided into eight evenly log-spaced bins. Cells were sorted into bins in proportion to the relative abundance of the library in each bin, with the most populous bin set to collect 3 million cells and around 0.5 million cells in the single-input and fine-tuning libraries (Fig. 2d and Extended Data Fig. 6d), respectively. For the multi-input base and expansion libraries, cells were split and either grown in DMEM containing 1 µM 4-OHT (Sigma-Aldrich), 20 µM GZV (MedChemExpress), 1 µM 4-OHT and 20 µM GZV or without inducers for 72 h (Supplementary Figs. 27 and 28). eGFP expression distributions for each condition were

then divided into eight evenly log-spaced bins and sorted in proportion to abundance, with the most populous bin set to collect around 1.5 million cells and around 1 million cells for the base and expansion libraries, respectively. A representative example of the flow sorting and BFP gating strategy used for all library sorting experiments is shown in Supplementary Fig. 38. For all libraries, the sorted cells were plated into 96-, 24- or 6-well plates to achieve a plating density of 10–30%, grown under puromycin selection for 2 days, passaged and washed with PBS, and grown for an additional 3 d. Cells were lifted using TrypLE and total RNA was extracted using the Takara RNA plus kit according to the manufacturer's instructions (Takara). Then, 5 μ l of total mRNA was converted to total cDNA using random hexamers according to the manufacturer's instructions (Verso, Life Technologies).

Short-read (Illumina) sequencing

To prepare the sorted libraries for short-read sequencing, barcode regions were PCR amplified from cDNA in a 15 μ l reaction with Phantamax polymerase (Vazyme) and custom bin-specific primer sets (IDT) using the following thermocycling protocol: 2 min at 98 °C, then 12–18 cycles (depending on the library) of 98 °C for 20 s, 60 °C for 10 s, 66 °C for 10 s, 72 °C for 20 s and then a final step of 72 °C for 2 min. The resulting amplicon pool was extracted from a 2% agarose gel. A second PCR step, which used 6–10 cycles (library dependent) of the thermocycling protocol for PCR step 1, added Illumina sequencing adapters (i5 and i7) and sample-specific sequencing barcodes using Phantamax and custom primer sets (IDT), and the amplified product was extracted from a 2% agarose gel. All primer sets used for NGS are reported in Supplementary Table 1. Purified amplicons from each bin were pooled at equimolar concentrations and sequenced using the Illumina MiSeq (kit v2, 300 cycles), NovaSeq6000 (Sp v1.5 kit, 300 cycles) and the NextSeq 2000 (P2 kit, 200 cycles) systems for the 384-, 166,000-, and 3.4×10^9 -member (both base and expansion) libraries, respectively. Illumina data were basecalled using the DRAGEN suite on basespace (Illumina), and further analysed using a custom analysis pipeline (MATLAB; Code availability and Supplementary Fig. 6). In brief, fastq files are imported and split according to their sample-specific sequencing barcode. Barcode sequencing data are converted to average expression for all compositions/variants as follows: (1) counts for each barcode in each sample/bin are normalized to the percentage of the library distribution in that bin to account for differences in read depth across bins. (2) A weighted average of number of reads in each bin for each barcode is then calculated to assign an average expression level (using a bin to expression conversion table) to each barcode. (3) Barcode expression levels are then cross-referenced with the Nanopore indices to assign average expression values for all barcodes associated with a given variant in a n -by- y cell array, where n is the number of library variants and y is the (variable) number of barcodes for each variant. (4) For each variant, a Gaussian kernel density estimation is performed over the log-normalized expression values from all barcodes associated with each variant, using the MATLAB function `ksdensity`. All local maxima that are not part of the dominant peak were suppressed, and expression values corresponding to the largest unimodal kernel density peak were assigned as the mean expression value for each variant. For visualization of the reconstructed histograms, secondary peaks are omitted.

Experimental validation of CLASSIC measurements

To assess the accuracy of CLASSIC measurements for both the EU and single-input libraries, single cells were sorted from the integrated cell populations into 96-well plates (MA900) and clonally expanded from across the mRuby and eGFP expression distributions, respectively. Isolated clonal populations, which are referred to as isolates, were expanded to 24-well plates and the fluorescence distribution was measured using flow cytometry (MA900). For the EU library, one 96-well plate of cells was sorted based on mRuby expression. For the

single-input base library, two 96-well plates of cells were sorted based on eGFP expression from the uninduced cell library. For the single-input fine-tuning library, one 96-well plate of cells was sorted based on eGFP expression from the uninduced population. For both the multi-input base and expansion libraries, two 96-well plates of cells were sorted based on eGFP expression from the uninduced cell libraries. After fluorescence measurement under library-specific experimental conditions, gDNA was extracted from each clonal population (Omega Bio-Tek). For the EU and single-input libraries, the genetic variant in each isolate was identified by PCR-based amplification and Nanopore sequencing of the LP locus using a forward primer in puromycin resistance cassette and a reverse primer in the BFP ORF (Supplementary Table 1). A set of 30 different forward primers, each with defined barcodes, were paired with a common reverse primer to amplify the genetic variants from each isolate and associate the genetic variant with the isolate during Nanopore sequencing. Then, 50 ng of gDNA was used as the template in a 20 μ l reaction along with 10 μ l of 2 $\times$ PhantaMax Master Mix (Vazyme), 1 μ l of each 10 μ M primer and brought up to 20 μ l in nuclease-free H₂O. The thermocycling protocol was 2 min at 98 °C, then 30 cycles of 98 °C for 20 s, 65 °C for 20 s, 72 °C for 5 min, followed by a final step of 72 °C for 10 min. Each PCR reaction was extracted from a 1% agarose gel and column-purified using QG buffer (Qiagen) according to the manufacturer's instructions. Purified amplicons were pooled and prepared for Nanopore sequencing using LSK-112 or LSK-114 ligation sequencing kit (ONT) and sequenced with the R10 (FLO-MIN112) or R10.4.1 (FLO-MIN114) MinION flow cell. Basecalling was performed using Guppy (v.6.0.0, and 6.5.7) or Dorado (ONT, super-high-accuracy mode) running on a GPU (NVIDIA RTX 3090). Composition and barcode assignment was performed using WIMPY, as described above. This process yielded 15 isolates from the EU library (Supplementary Fig. 12b), 40 isolates from the single-input base library (Fig. 2d (middle), Supplementary Fig. 35a and Supplementary Table 2) and 8 isolates from the single-input fine-tuning library (Extended Data Fig. 6f and Supplementary Fig. 35d). For isolates from the base and expansion multi-input libraries, the circuit identity was determined by amplification of the barcode sequence from gDNA, following the PCR protocol described above, extraction from a 2% agarose gel and Sanger sequencing (the primer sequences are provided in Supplementary Table 2). Using the CLASSIC composition-to-barcode IDs to assign barcodes to circuit variants yielded 14 and 13 mapped isolates from the base and expansion libraries, respectively (Supplementary Figs. 29 and 30 and Supplementary Table 2).

MLP deep neural network regression for circuit behaviour prediction

Model architecture and parameters. Fully connected MLP neural network architectures were constructed for single- and multi-input design spaces. The single-input model had four hidden layers with 160/80/40/20 neurons, a 39×1 input channel, and 2 output channels for log₁₀-transformed basal and induced expression. The multi-input model consisted of 5 layers 160/80/40/20/10 neurons, a 68×1 input channel and 4 output channels. Input compositions were one-hot encoded for each part category and flattened into a 39×1 shape for the single-input library and a 68×1 shape for the multi-input library. Stochastic gradient descent with momentum (sgdm) was used to optimize the root mean squared error loss function during training (momentum parameter = 0.9), learning rate of 5×10^{-2} , and a piecewise learn rate scheduler that drops the learning rate by a factor of 0.5 every 10 epochs. The batch size was 256.

Data splits for model training. High-quality test sets were constructed from the single- and multi-input base library datasets (121,292 and 88,551 measured compositions, respectively), by randomly sampling 15% of compositions with >12 and >50 barcode-indexed reads (1,868 compositions and 953 compositions respectively). Flow cytometry

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measurements for isolates were removed from training datasets to allow for unbiased comparisons against these measurements later. The remaining data were randomly split 90:10 into training/validation sets containing 103,878 compositions and validation sets containing 11,532 compositions for the single-input library, and 48,031 and 5,336, respectively, for multi-input library. The validation set was used to monitor training progress (Extended Data Figs. 3 and 7), and for early stopping with a patience parameter of 300 iterations. The model with the best validation performance was returned and used to evaluate performance on the high-quality test and isolate sets.

Data downsampling. For demonstrating the single-input model's tolerance to lower amounts of training data (Fig. 3f), compositions were downsampled in increasing order of the number of barcoded reads per composition, retaining compositions from ≥ 1 read to ≥ 40 reads to simulate scenarios in which lower amounts of higher-quality data are collected. The model was reinitialized and trained with a new 90:10 training-validation split from the remaining dataset for each of the 40 downsampling iterations, which were all evaluated using the high quality test set mentioned above.

Design space expansion through neural network fine-tuning. To demonstrate single-input model fine-tuning, an MLP model with the same architecture and hyperparameters was trained on the original library, with two extra input encoding slots to represent new parts that would be added during fine-tuning. For the multi-input library, the base model already contained extra input slots for new parts. A high-quality test set was created, as described above, from the expansion library datasets, with an increased barcoded read count threshold of 40 (single-input) or 50 (multi-input). The remaining data were split 90:10 into a training and validation temp sets. These sets were combined with 2,000 compositions from the base training set and 500 compositions from the base validation set to avoid memory loss during training (Extended Data Fig. 6g,h). The combined validation set (with base and new compositions) was used for early stopping and all hyperparameter optimization as described above, except for the initial learning rate, which was changed to 5×10^{-3} . The model performance was evaluated on the high-quality test sets and used to construct individual variants across the behaviour space.

MI calculation

For a given pair of features in a region of design space, MI was computed using the following formula:

$$\sum_{x,y} p(x,y) \log_2 \frac{p(x,y)}{p(x)p(y)}$$

where x and y are 2 different features of the library, and $p(x,y)$ represents the joint frequency for specific x and y combinations divided by the total number of points in the dataset. $p(x)$ and $p(y)$ represent the fractional frequency of the corresponding parts by themselves.

SHAP and UMAP clustering

For both the single- and multi-input design spaces, SHAP values were used to assess the contribution of each feature in the HFC and AND/OR-like regions of behaviour space, respectively. For each model, SHAP values were computed by fitting a SHAP explainer model across 500 randomly sampled training points and then predicting on 200 randomly sampled points sampled from the regions of interest. These values were averaged across expression conditions and within part categories to generate region-specific SHAP scores for each part category.

For the 166,000-member library, UMAP dimensional reduction was performed on the ten categorical part encodings for part composition data from variants in the HFC (>25-fold region) region of the behaviour space, using the `run_umap` function in MATLAB with a Jaccard

distance metric. To identify the optimal number of clusters, a gap test was carried out on the 2D projection using the `cluster_evaluate` function in MATLAB. HFC compositions were assigned to clusters using the `kmeans` function in MATLAB. Cluster stability was determined by performing 100 independent rounds of UMAP projection and k -means clustering, recording cluster affiliation in each iteration, and computing the cluster stability index and adjusted rand index. We then assigned compositions to clusters the most stable UMAP projection from the 100 runs mentioned above. Clustering and computation of the cluster stability for AND-like and OR-like circuits from the multi-input library was performed as described above for the single-input library, but on the 15 categorical part encodings in either region. For display purposes, boundaries have been manually drawn around each cluster.

KL-divergence computation

For multi-input circuits, expression levels were normalized to relative abundances by dividing them by the summed expression across the four induction conditions (no inducer/4-OHT/GZV/both) creating probability distributions suitable for computing Kullback-Leibler divergence (D_{KL}) against ideal AND and OR gates, using the following equation:

$$D_{KL} = \sum p(x) \cdot \ln \left(\frac{p(x)}{q(x)} \right)$$

Where $p(x)$ represents the normalized distribution across the four conditions, and $q(x)$ refers to the probability of the ideal AND [0.01, 0.01, 0.01, 0.97] and OR [0.01, 0.33, 0.33, 0.33] gate across the four induction conditions. Low D_{KL} values indicate high similarity to ideal AND and OR distributions. Crucially, high D_{KL} values can result from a wide variety of distributions, whereas low D_{KL} values reflect a restricted and highly specific set of behaviours closely resembling the ideal AND or OR gate.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All Nanopore and Illumina sequencing datasets generated in this study are available from the Sequencing Read Archive (BioProject: PRJNA1347054).

Code availability

All custom scripts used for Nanopore sequencing data analysis are available at GitHub (<https://github.com/cbashorlab/WIMPY>). Code associated with Illumina data analysis and model training are available at GitHub (<https://github.com/cbashorlab/CLASSIC>). All other scripts used to generate any analysis in addition to those provided above are available on request.

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Author contributions R.W.O., K.R. and C.J.B. conceived the study. R.W.O. and K.R. carried out the experiments and developed the analysis software, with assistance from T.C.P., Y.W., L.B.C.B., K.D.S., J.A.W., S.L., T.H.Z., E.M.R. and A.S.; R.W.O., K.R. and T.C.P. developed the modular

cloning scheme and LP cell line, with assistance from S.L.; B.K., K.D.C. and T.J.T. helped to develop the barcoding scheme and analysis software. R.W.O., K.R., T.C.P., Y.W., J.W.R., P.M. and C.J.B. analysed the data. C.J.B. supervised the study. R.W.O., K.R. and C.J.B. wrote the manuscript, with input from all of the authors.

Competing interests A provisional patent application that covers technologies described in this Article has been filed by Rice University.

Additional information

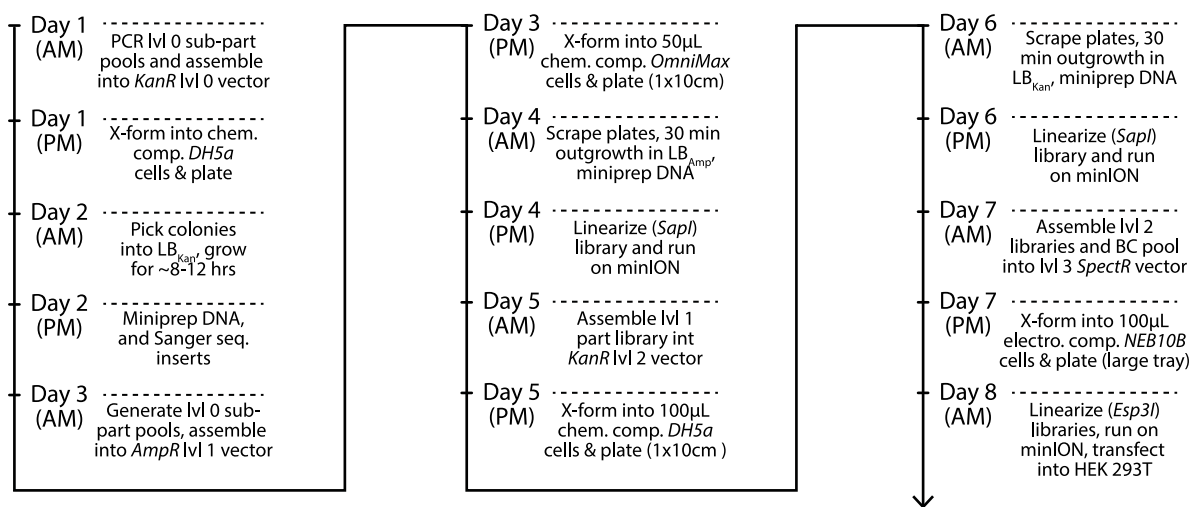
Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-025-09933-9>.

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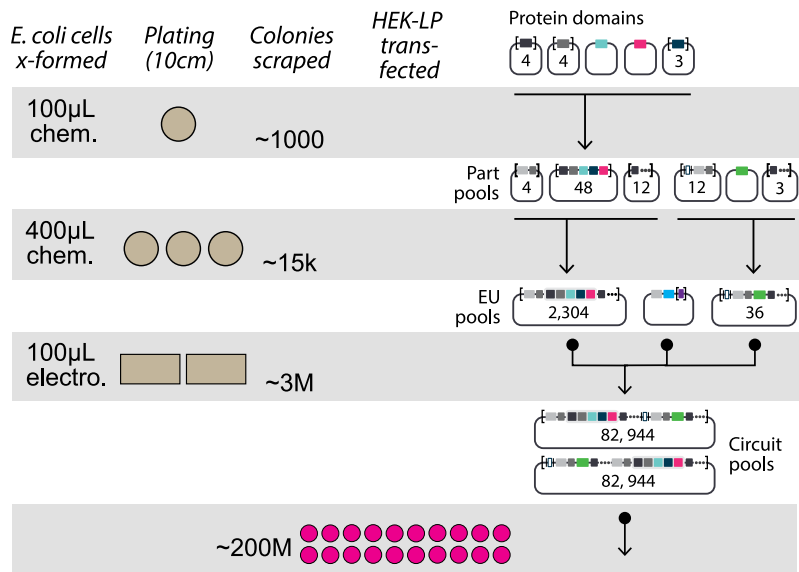
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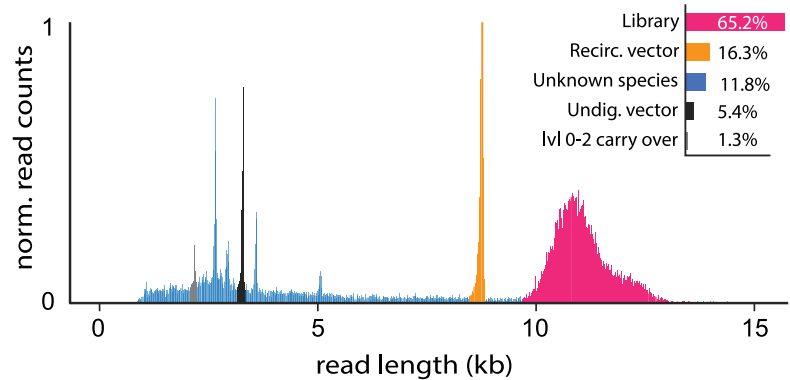
a 166K-LIBRARY CLONING PIPELINE



b EXPERIMENTAL OVERVIEW



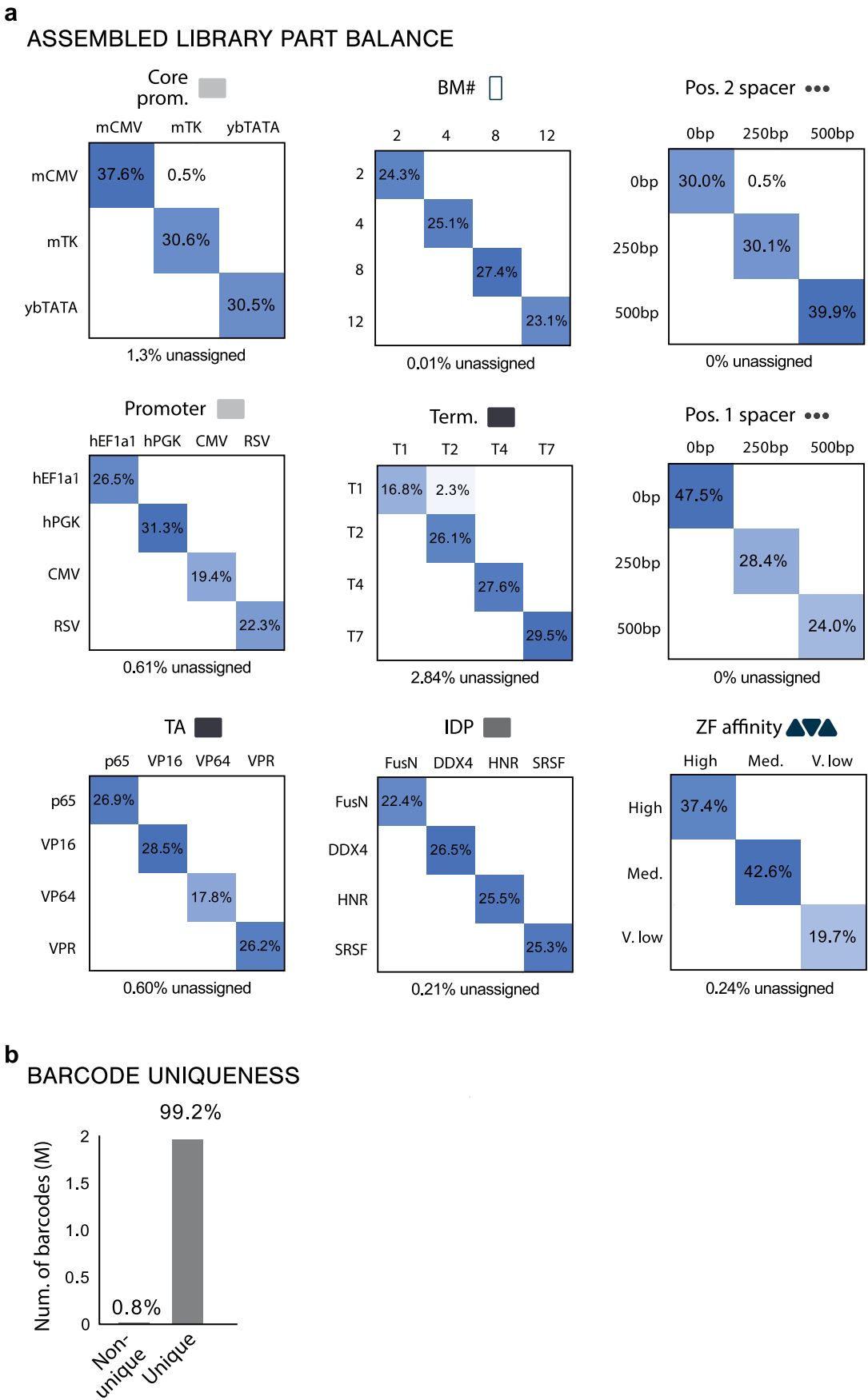
c NANOPORE ANALYSIS OF LIBRARY ASSEMBLY



Extended Data Fig. 1 | See next page for caption.

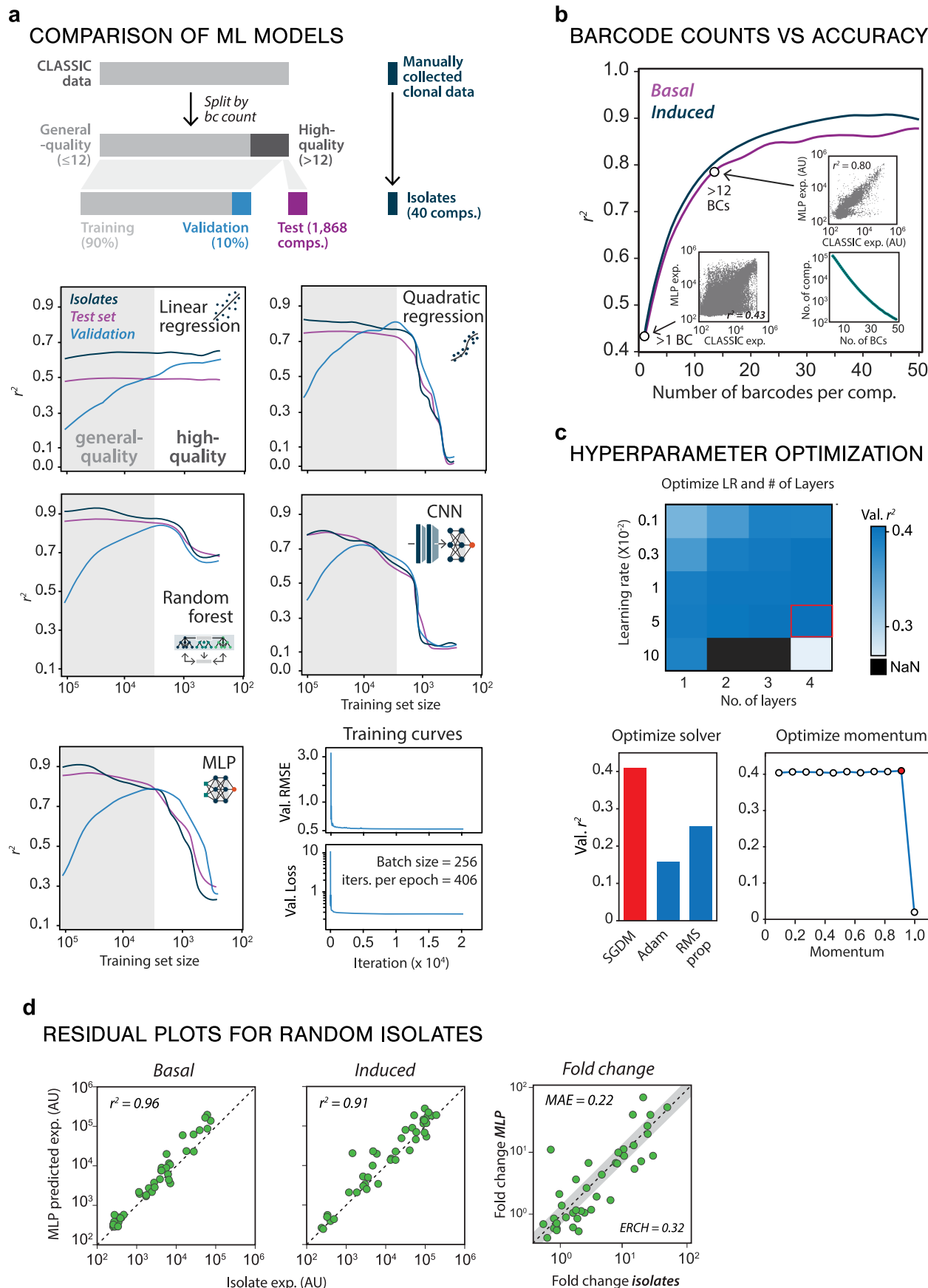
Extended Data Fig. 1 | Overview of 166K-member gene circuit library construction. **a**, Timeline of library construction process from level 0 part fragments to level 3 libraries. **b**, Schematic of the construction strategy for each level of the assembly process (grey boxes), with number of cells indicated for assembly transformation and plating (brown circles, 10 cm plates; brown rectangles, autoclave trays), colony scraping, and HEK-LP transfection (pink circles, 10 cm plates). The inputs and products for each assembly level

are represented to the right. **c**, Nanopore sequencing read-length distribution of level 3 single-input inducible circuit library, with the relative proportion of each identifiable DNA product denoted: library members (pink), re-circularized level 3 destination vector (orange), empty level 3 destination vector (dark grey), contamination from level 0, 1, or 2 genetic parts (light grey), or unidentified DNA species (blue).



Extended Data Fig. 2 | See next page for caption.

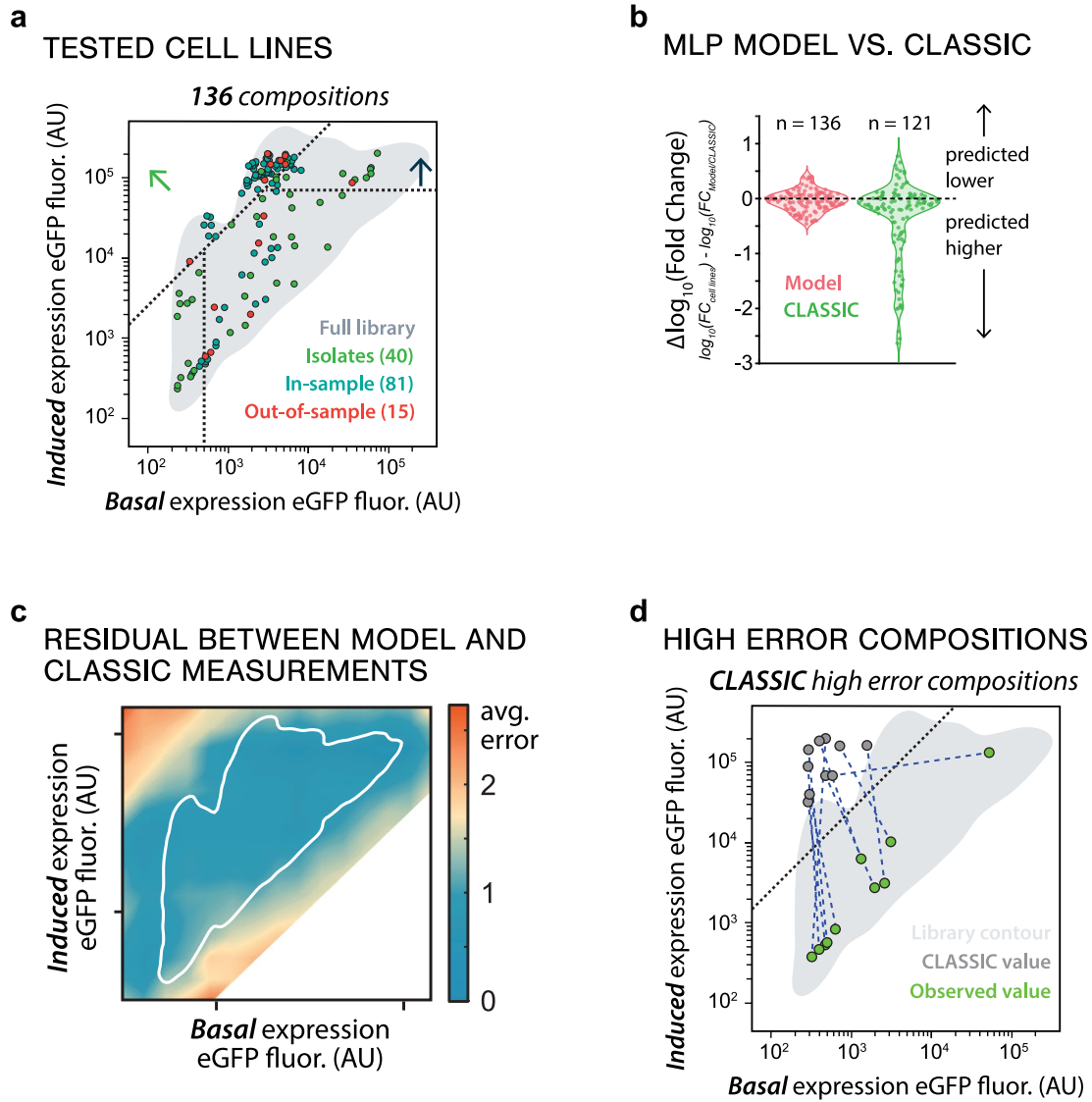
Extended Data Fig. 2 | 166k-member library balance and barcoding analysis metrics. **a**, Confusion matrices showing the percentage of reads unambiguously assigned to each genetic part (on-diagonal) and reads ambiguously linked to two parts (off-diagonal) following level 3 library assembly. Individual values representing <0.5% of the total reads are not shown. The percentage of reads in which a part identity could not be determined are shown at the bottom of the respective part confusion matrices. **b**, Number of barcodes determined to be uniquely mapped to a single composition (“unique”) or multiple compositions (“non-unique”), as determined from Nanopore sequencing analysis of the level 3 library.



Extended Data Fig. 3 | See next page for caption.

Extended Data Fig. 3 | Comparison and optimization of ML models. **a.** Top: schematic outlining the process of generating training (light grey), validation (blue), test (purple), and isolate (navy) sets. CLASSIC data sets are first divided into high- (>12 barcodes, dark grey) or general-quality (<12 barcodes, light grey) sets before generating the training, validation, and test splits. Bottom: comparison of the performance of 5 different model classes (linear regression, quadratic regression, random forest (RF), convolutional neural network (CNN), and multi-layer perceptron (MLP)) for predicting circuit behaviour using varying amounts of training data (x-axis), as monitored using test (purple line), validation (light blue line), and isolate (navy line) set r^2 values (y-axis) (see supplementary text section S3.5). Grey shaded region on each plot represents a regime in which the training set is dominated by general-quality reads. For the selected model class (MLP), training curves of the root mean squared error (RMSE) and loss for the validation set are provided. **b.** Basal (purple line) and induced (navy line)

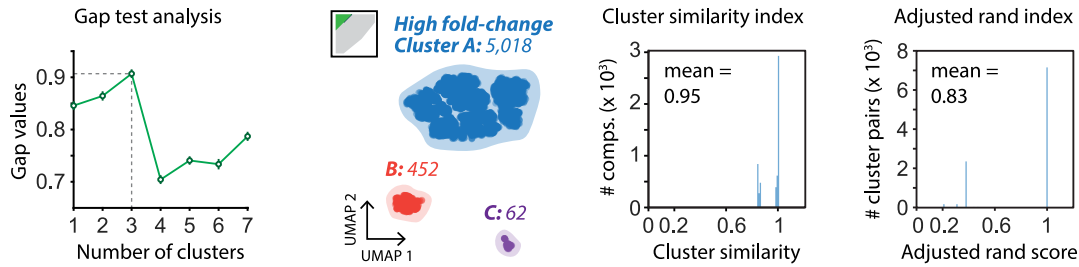
r^2 values for predicted vs observed CLASSIC expression from the trained MLP. Insets represent CLASSIC vs predicted measurements for > 1 (bottom left, $r^2 = 0.43$) and > 12 (top middle, $r^2 = 0.80$) barcodes per composition, and the number of compositions for increasing number of barcodes per composition (bottom right). **c.** Hyperparameter optimization (HPO) of the MLP, monitored using the validation set: learning rate (LR, y-axis) for different numbers of layers (x-axis) with 4 layers and a learning rate of 5×10^{-2} providing highest r^2 (red outline) (top); solver choice, with SGDM leading to the highest r^2 (red bar) (bottom left); momentum, with a momentum of 0.9 providing the highest r^2 (red circle) (bottom right). **d.** Comparing ground-truth basal expression (left), induced expression (middle), and fold change measurements (right) (x-axis) with HPO MLP predictions (y-axis) for the isolate set ($r^2 = 0.96$, $r^2 = 0.91$, MAE = 0.22, respectively, $n = 40$).



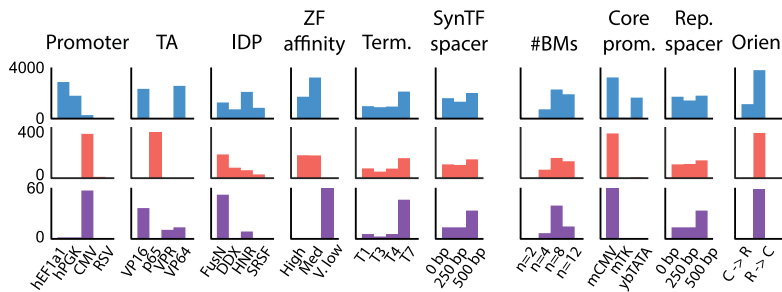
Extended Data Fig. 4 | Summary of cell lines constructed from the 166k-library design space to validate predictions from the MLP model. a, Basal and induced eGFP expression levels for each constructed cell line circuit composition overlaid onto single-input behaviour space (clonally isolated, green; constructed out-of-sample, red; constructed in-sample, teal; contour for 97.5% of compositions in the MLP-predicted behaviour space, grey). Dotted lines separate behavioural regions of interest: low basal (<500 AU), purple arrow; high induction (>70k AU) blue; high fold-change (HFC) (>25x, green). **b,** Comparing fold-change values for MLP model-predictions (red, left, n = 136) or CLASSIC

measurements (green, right, n = 121) with ground truth cell line measurements. **c,** Residual between model and CLASSIC measurements. Heatmap corresponds to the manhattan distance between residuals of basal and induced expression values calculated across a 20×20 grid in the behaviour space. Variants were assigned to grids based on CLASSIC measured values. **d,** Basal and induced expression behaviour for the top 10 highest error compositions from the individual variants shown in (a). CLASSIC measurements (grey) and corresponding ground truth values measured from constructed cell lines (green) are shown. Lines (blue dashed) link each pair of CLASSIC and ground truth expression.

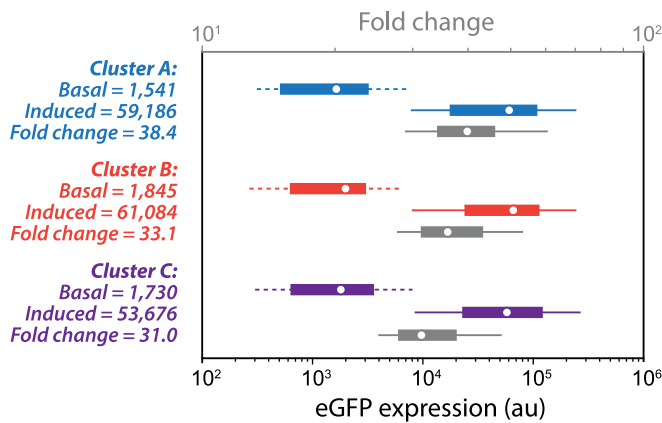
a CLUSTER STABILITY ANALYSIS



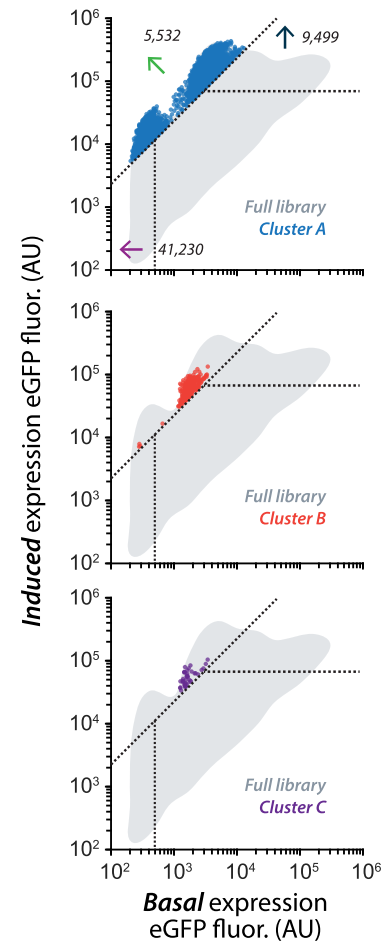
b HFC CIRCUIT VARIANTS PART USAGE



d EXPRESSION BEHAVIOR ACROSS CLUSTERS



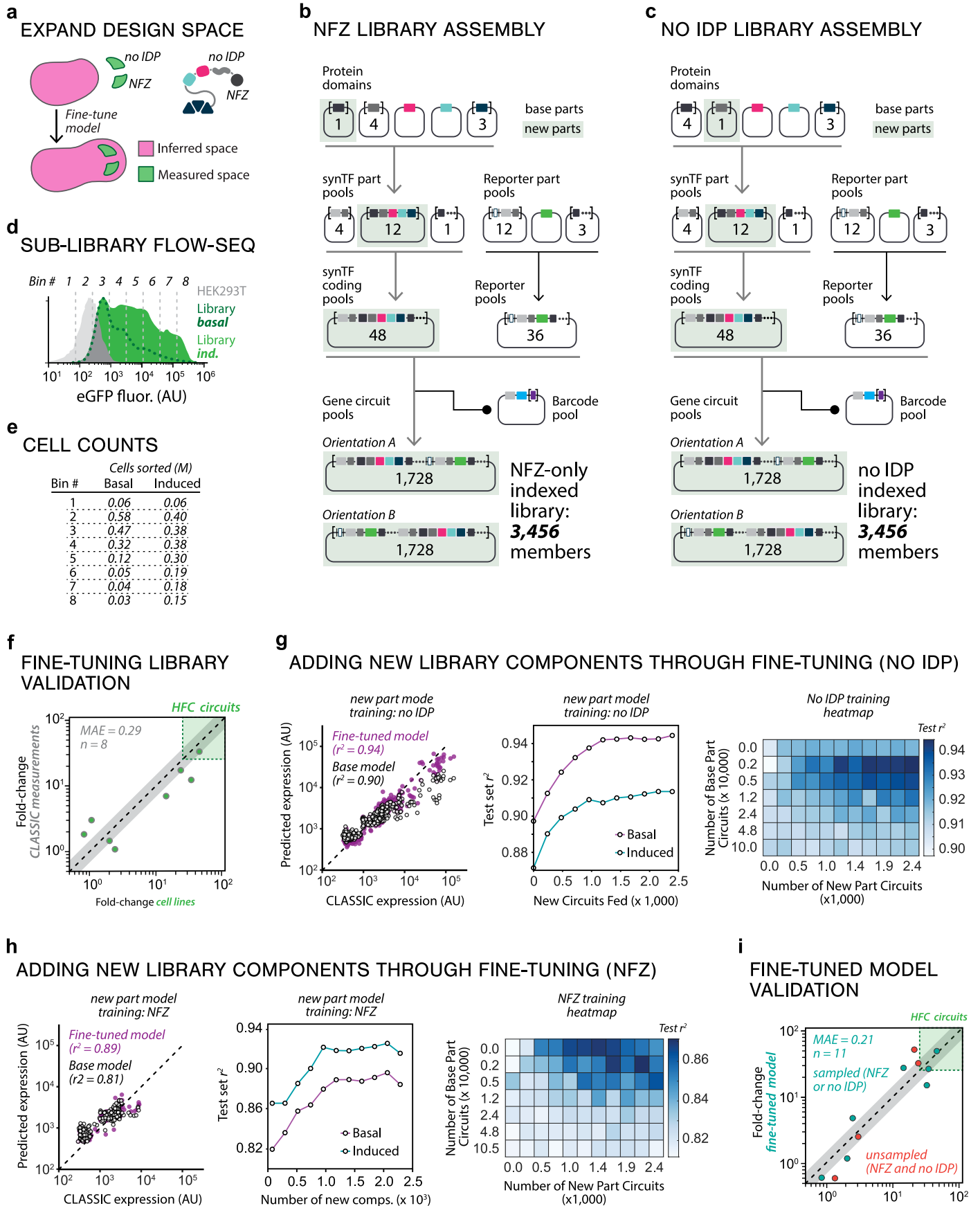
c CLUSTER DISTRIBUTIONS



Extended Data Fig. 5 | Clustering analysis of HFC compositions. **a**, Gap test for cluster number (left) and subsequent UMAP projection of HFC variants grouped into 3 cluster (cluster A, blue; cluster B, red; cluster C, purple) (middle left). Cluster similarity scores from 100 independent k-means clustering outcomes for all compositions, and adjusted rand score for all pairs of clustering results (right). Means represent the population cluster similarity and adjusted rand index respectively. **b**, Part usage frequency for variants in each cluster. **c**, Mapping of basal and induced eGFP expression of compositions from each

of the three clusters, overlaid on a contour constructed from 97.5% of the data from the MLP-predicted behaviour space (see Fig. 4a) (grey fill). **d**, Distribution of basal (dotted line) and induced eGFP expression (solid line) (bottom axis), as well as fold change values (grey line, top axis) for each cluster. Circles represent the median values, boxes span the 25th to 75th percentiles, and the upper and lower whiskers represent the median $\pm 1.5 \times$ IQR (line ends). Median values are shown to the left of the plot. Sample sizes (n): Cluster A = 5,018, Cluster B = 452, and Cluster C = 62.

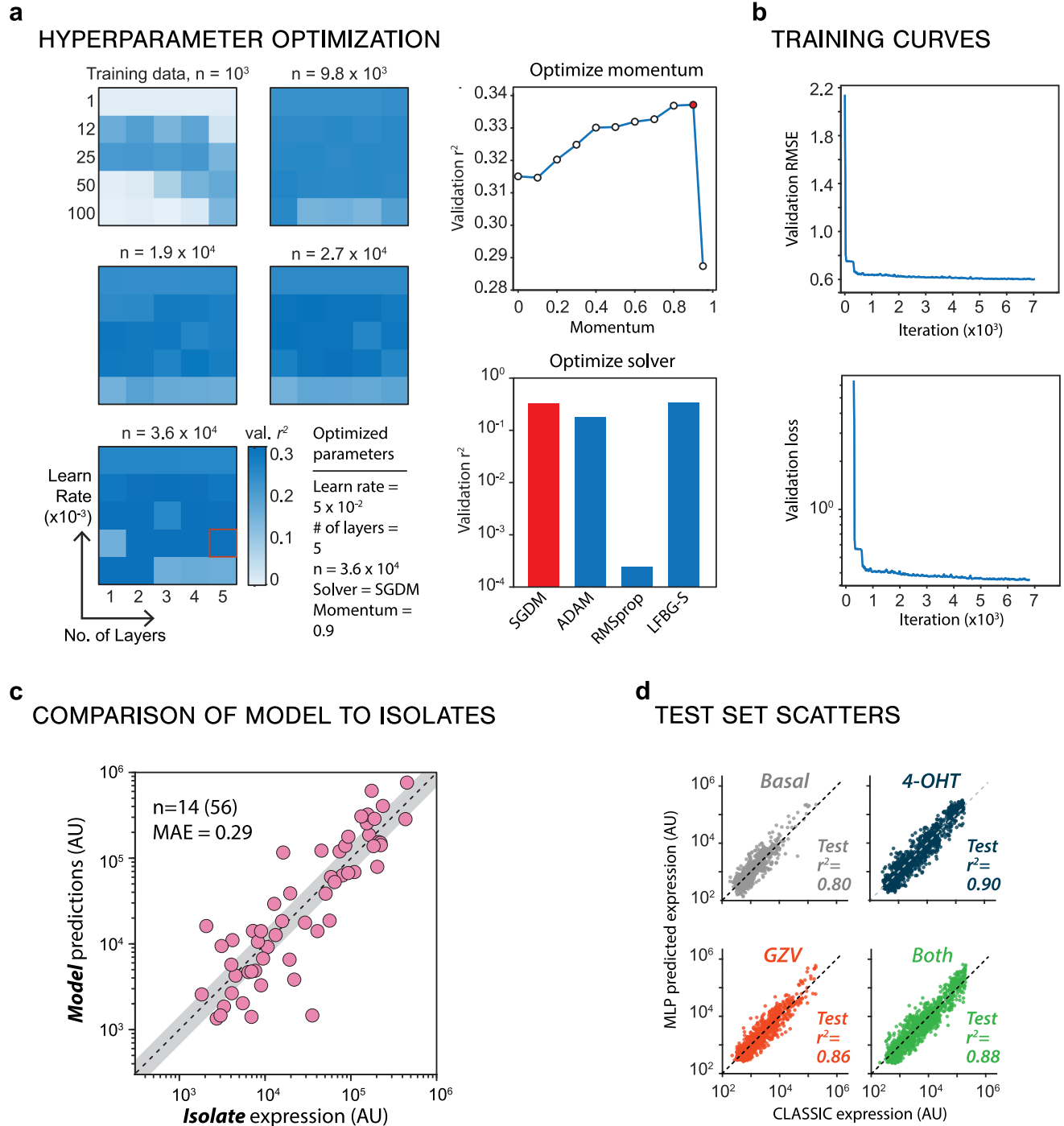
Article



Extended Data Fig. 6 | See next page for caption.

Extended Data Fig. 6 | Fine-tuning model. **a**, Schematic depicting a proposed method for expanding the model-predicted design space (red) to include 2 new parts (NFZ and no IDP) by fine-tuning the model using small libraries of new parts (green) (left). Representation of the position of the new parts in the synTF architecture (right). **b**, Schematic outlining the assembly strategy to retroactively add the TA NFZ to the design space. Transparent green boxes signify individual plasmids or plasmid pools that contain the new part. **c**, Schematic outlining the assembly strategy to retroactively add IDP-less variants the design space. Transparent green boxes signify individual plasmids or plasmid pools that contain the new part. **d**, eGFP distributions for the new libraries to explore this sub-space. **e**, Table of the number of cells sorted into each bin for both inducer conditions during flowSeq. **f**, 8 individually constructed variants from the sub-space to validate CLASSIC measurements. Grey region, ERCH; Green square,

HFC region. **g & h**, Comparison of basal eGFP expression predictions and CLASSIC measurements for a high-quality test set of compositions lacking an IDP (panel **g**) or containing an NFZ TA (panel **h**), using either a base model (white dots with black outline, $r^2 = 0.90$ or $r^2 = 0.81$, respectively) or a fine-tuned model (purple dots, $r^2 = 0.94$ or $r^2 = 0.89$, respectively) (left). Breakdown of the basal (purple) and induced (teal) expression prediction accuracy with increasing amounts of fine-tuning data from the IDP lacking (**g**) or NFZ-containing (**h**) libraries, as assessed by monitoring the test set r^2 (middle). A 2D map outlining the amounts of base library and no IDP library (**g**) or NFZ-containing (**h**) data required for optimal fine-tuning of the base model, as determined by the test set r^2 (right). **i**, 11 individually constructed variants from the sampled (teal) and un-sampled (red) expanded design space to validate fine-tuned model predictions. Grey region, ERCH; Green square, HFC region.

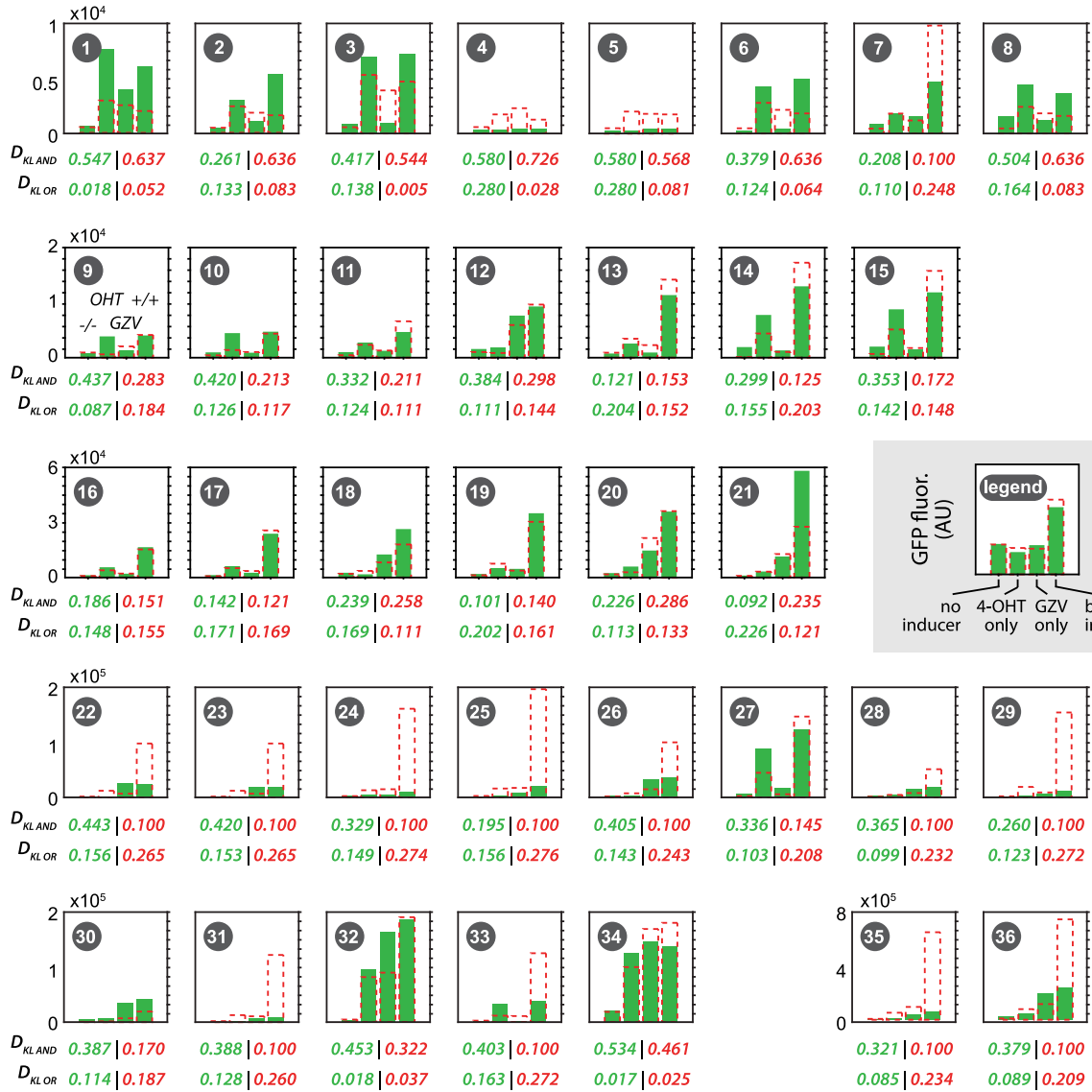


Extended Data Fig. 7 | Hyperparameter tuning and validation of base MLP model. **a**, Hyperparameter optimization for the multi-layer fully connected neural network. Validation r^2 values for (left) 2D combinations of learning rate (y-axis) and number of layers (x-axis) for varied amounts of training data used, (top right) momentum parameter for stochastic gradient descent with momentum (SGDM), and (bottom right) different solvers. Most optimal parameter from each scan is shown in red. **b**, Training curves showing RMSE

(top) and loss (bottom) as a function of training iteration for the validation set. Training was stopped with a validation patience parameter of 300 iterations. **c**, Comparison of model predictions to randomly isolated clones from the library. MAE: mean absolute error. **d**, Scatter plots of the test set from the base model for each of the four input conditions (basal: light grey, navy: OHT only, orange: GZV only, green: both inducers). r^2 , Pearson's r^2 .

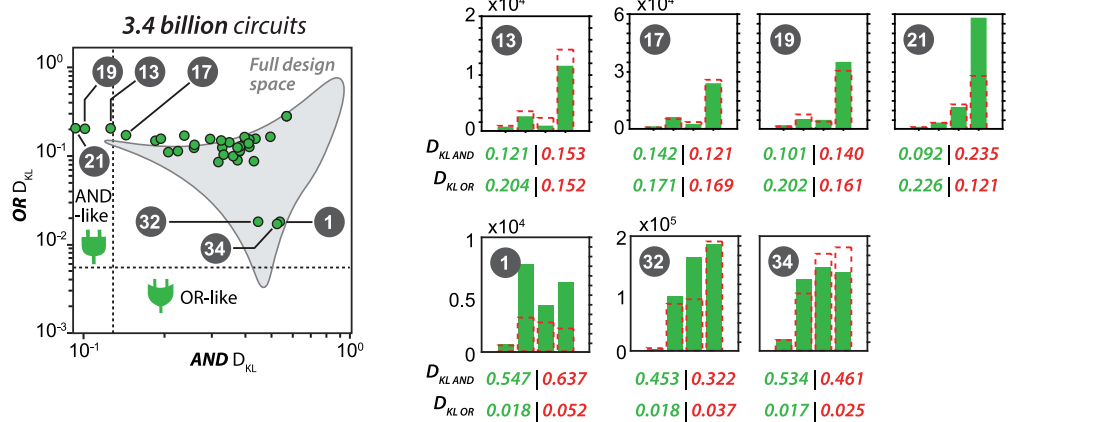
a

COMPARISON OF MODEL TO INDIVIDUALLY CONSTRUCTED CELL LINES



b

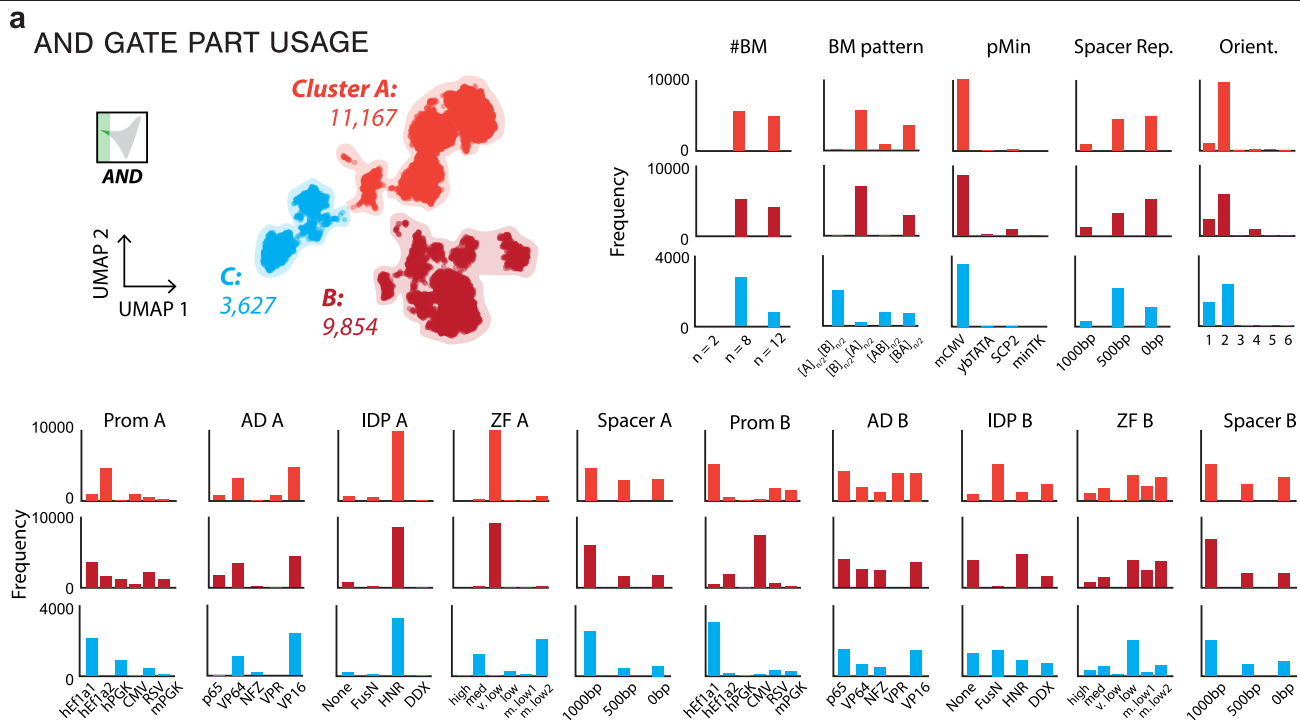
INDIVIDUALLY CONSTRUCTED CELL LINES PLOTTED IN AND-OR COORDINATE SPACE



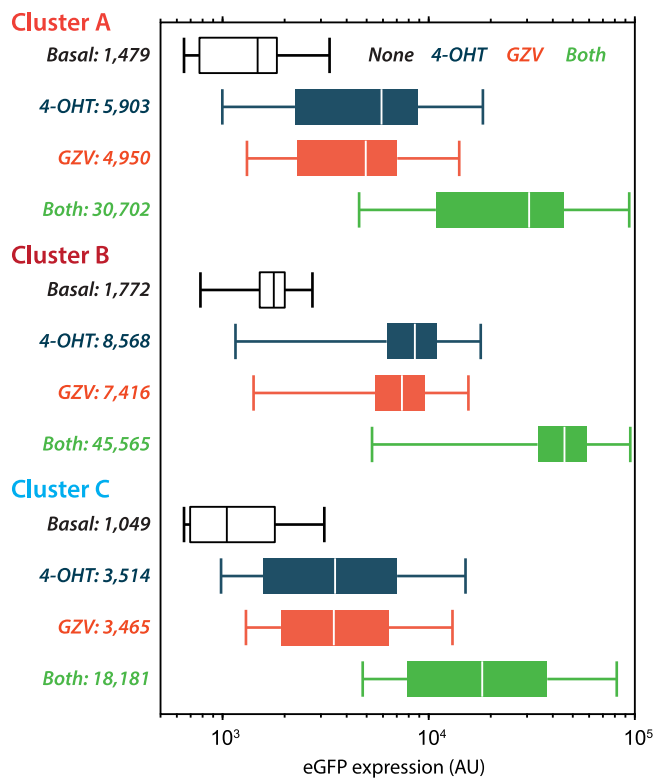
Extended Data Fig. 8 | See next page for caption.

Extended Data Fig. 8 | Validation of model predictions with individual measurements. **a**, Flow cytometry was used to measure basal, OHT-induced, GZV-induced, and dual-induced eGFP expression levels for 36 individually constructed cell lines harbouring integrated circuit compositions sampled from across the multi-input library behaviour space. Green bar plots (flow cytometry measurement) and dotted red outlines (model predictions) are shown for each circuit for all four conditions (far left, basal; middle left,

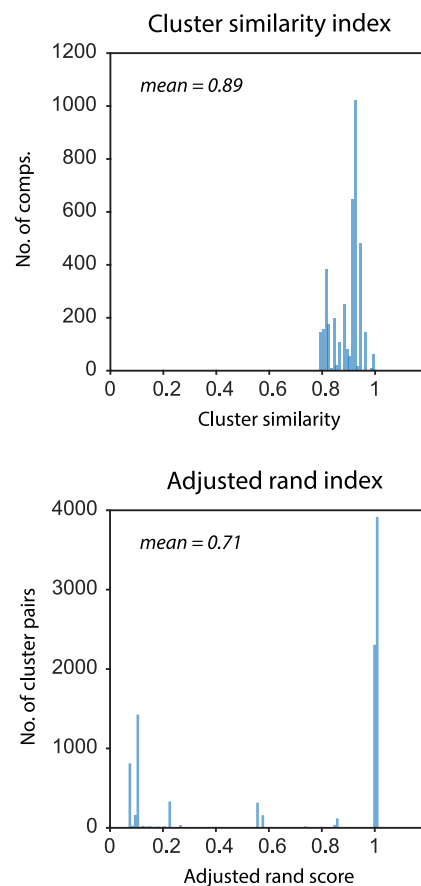
4-OHT-induction; middle right, GZV-induction; far right, dual induction). Bars represent the mean of the expression distribution for a single measurement. KL-divergence (D_{KL}) from AND (top) or OR (bottom) gate shown below each plot (prediction, red; measurement, green). Numbers in grey circles represent an index for that circuit. A legend explaining the layout of each plot is shown in a grey rectangle. **b**, The AND-OR coordinates of each cell line (green dots), superimposed on the contour of the design space (grey).



b AND GATE CLUSTER PERFORMANCE

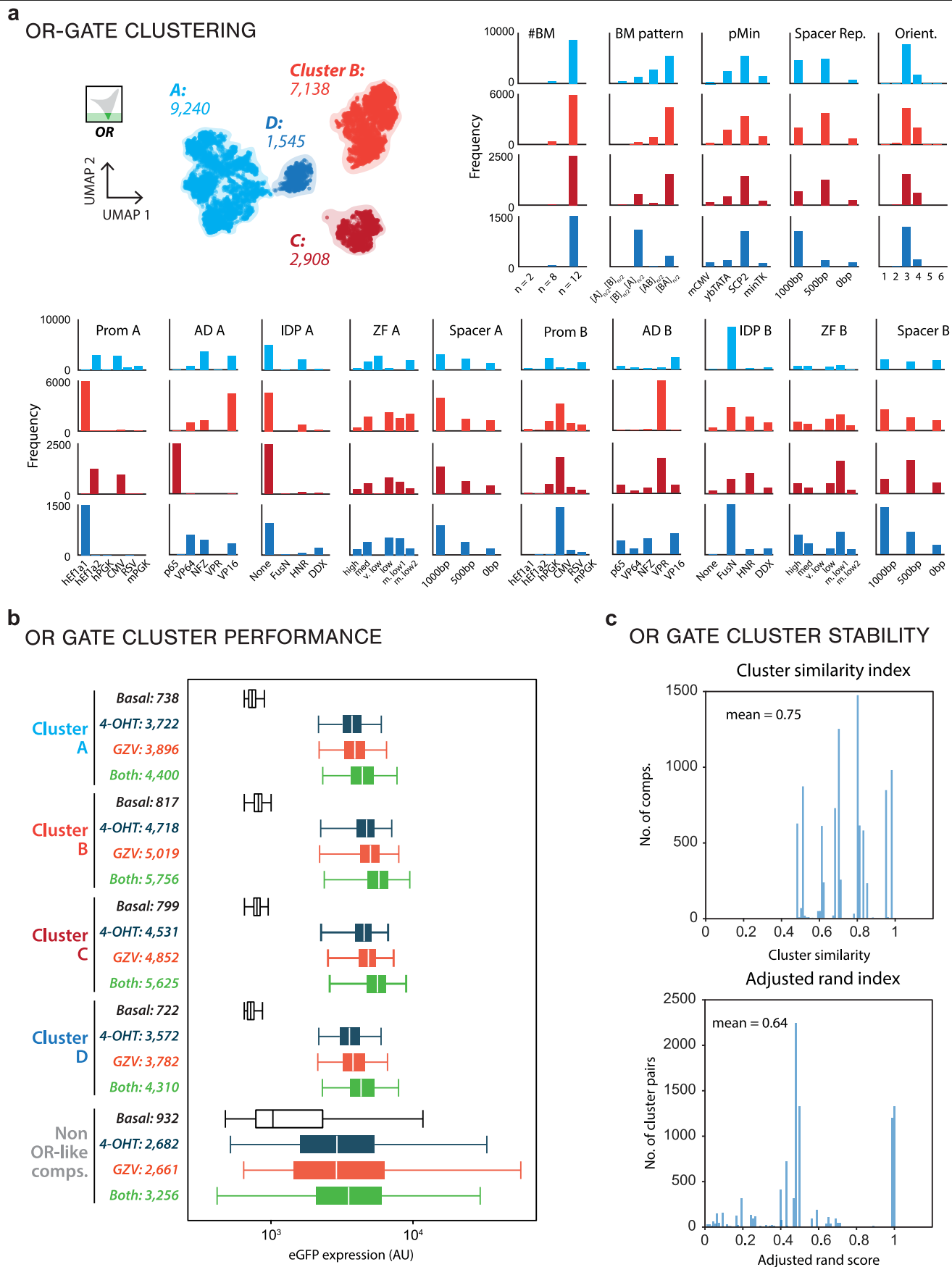


c AND GATE CLUSTER STABILITY



Extended Data Fig. 9 | Extended AND-gate clustering analysis of the multi-input library. **a**, Clustering of the multi-input AND-like behaviour space (top left) and part usage across the clusters. **b**, AND cluster expression distributions across the 4 input conditions (basal, black; 4-OHT, navy; GZV, orange; Both, green), represented by boxplots outlining the interquartile range (IQR) (box), the median (white band), and the median $\pm 1.5 \times$ IQR (line ends).

Median values are shown to the left of the plot. Sample sizes (n): Cluster A = 11,167, Cluster B = 9,854, and Cluster C = 3,627. **c**, Cluster stability analysis of compositions in the AND-like behaviour space by computing the cluster similarity index (top) across 100 UMAP projections and cluster calculations, and adjusted rand index (bottom) across every pairwise combination of clustering results across the 100 UMAP projections and cluster calculations.



Extended Data Fig. 10 | See next page for caption.

Extended Data Fig. 10 | Extended OR-gate clustering analysis of the multi-input library. **a**, Clustering of the multi-input OR-like behaviour space (top left) and part usage across the clusters. **b**, OR cluster expression distributions across the 4 input conditions (basal, black; 4- OHT, navy; GZV, orange; Both, green), represented by boxplots outlining the interquartile range (IQR) (box), the median (white band), and the median $\pm 1.5 \times$ IQR (line ends). Median values

are shown to the left of the plot. Sample sizes (n): Cluster A = 9,240, Cluster B = 7138, Cluster C = 2,908, and Cluster D = 1,545. **c**, Cluster stability analysis of compositions in the OR-like behaviour space by computing the cluster similarity index (top) across 100 UMAP projections and cluster calculations, and adjusted rand index (bottom) across every pairwise combination of clustering results across the 100 UMAP projections and cluster calculations.

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Flow cytometry data were collected using Sony MA900 Software. All Nanopore sequencing data were collected using minKNOW and basecalled using Guppy and Dorado software packages (version 6.0.0 - 6.5.7, as appropriately available on https://nanoporetech.com/software/other/guppy/history) from ONT. All Illumina sequencing data were collected and basecalled using Basespace and the DRAGEN suite.
Data analysis	Flow cytometry data were analyzed using FCS Express 6. All Nanopore and Illumina sequencing data were analyzed using custom scripts developed in Matlab 2022a & 2024a (detailed description in supplementary information). Custom scripts developed for analysis of nanopore data are available on the github repository - github.com/cbashorlab/WIMPY . This includes the entire WIMPY software suite, including the functions fastqall, tilepin, bowtile, chophat, viscount, fastar and barcoat (no specific software version). All custom scripts used to analyze illumina data are provided on the github repository - github.com/cbashorlab/CLASSIC .

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

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All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All nanopore and Illumina sequencing datasets generated in this study are available from the Sequencing Read Archive (BioProject ID PRJNA1347054).

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

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Population characteristics

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Ethics oversight

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes for library construction were not predetermined. We aimed to get between 10 and 100x coverage of our library at each of the stages of the CLASSIC process: construction, genomic integration, and phenotypic selection. Sample sizes for validation experiments were determined by balancing practical constraints with the need for comprehensive behavior space coverage. We constructed a minimum of 8-40 individual variants for each library validation, strategically sampling across the full range of predicted behaviors rather than random sampling. We chose mean absolute error (MAE) as our primary validation metric because it provides a direct measure of prediction accuracy that is robust to outliers and doesn't require the distributional assumptions of correlation-based metrics. Unlike Pearson's correlation, which can be misleading with small samples or non-linear relationships, MAE gives an interpretable measure of how far off our predictions are in actual fluorescence units. The sufficiency of our sample sizes is demonstrated by: (1) the consistency of MAE values across different library validations (MAE ranging from 0.07-0.32), (2) the agreement between CLASSIC measurements and ground-truth isolates across multiple independent experiments, and (3) the successful experimental validation of model predictions for edge-case behaviors like high fold-change and Boolean logic gates.
Data exclusions	Barcode sequences corresponding to multiple different genetic compositions were excluded from analysis, as mentioned in supplementary information. This filtering step was a predetermined exclusion requirement that aimed to improve the accuracy of the measurements, as multiple compositions linked to a single barcode generate noisy, inaccurate measurements. No other data were excluded.
Replication	In order to validate reproducibility of our technique, we repeated construction, integration, sorting, and data analysis of a 384-member library (fig. S12C and D). Additionally, for each library we compared the behavior of CLASSIC measurements to individually constructed and measured library members (n=15 for 384-member; n = 40 for 166k-member; n = 27 for 3.4B-member). All attempts at replication were successful.
Randomization	Not relevant since data were quantitative and did not require subjective grouping.
Blinding	Not relevant since data were quantitative and did not require subjective grouping.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	HEK293T (ATCC, CRL-11268, Female), HEK-LP (derived from HEK293T as described in the Methods and Supplementary Text)
Authentication	HEK-LP cell line was authenticated by standard PCR and ddPCR and clones isolated from the library were authenticated by PCR of the integrated locus and subsequent Nanopore amplicon sequencing (as detailed in the Methods). No further authentication has been conducted. HEK293T cell line was not authenticated.
Mycoplasma contamination	Cell lines tested negative for mycoplasma contamination.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in this study.

Plants

Seed stocks	n/a
Novel plant genotypes	n/a
Authentication	n/a

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Cells were washed twice with PBS and resuspended in complete DMEM at a concentration of ~10M cells/mL
Instrument	Sony MA900

Software	Sony MA900 software, FCS Express 6
Cell population abundance	BFP and GFP or mRuby expression: the purity of post-sort fractions were confirmed by flow cytometry.
Gating strategy	Cells were gated on a constant BFP+ expression gate after gating for single cells. The BFP+ gate was determined using a negative control sample (HEK293T WT).

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.